

HYPERSONIC JET IMPINGEMENT DURING LANDING ON LUNAR AND MARTIAN SURFACES

Report on MEng. Final Year Internal Project

Jack Davies

University *of* Glasgow
School *of* Engineering

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Hypersonic Jet Impingement During Landing on Lunar and Martian Surfaces

Name of First Supervisor:

Dr Hossein Zare-Behtash

Name of Second Supervisor:

Dr Douglas Thomson

Student Name:

Jack Davies

Matriculation Number:

2391088D

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Aeronautical Engineering

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Abstract

A computational numerical method of simulating impinging hypersonic plume flows in a near-vacuum environment for both the near- and far-field flow regimes was created and validated. This work included the development of a thorough overview of contemporary knowledge of a fluid dynamic phenomenon known as Plume Surface Interactions (PSI), which describes the complex and non-linear interactions which occur between an expanding jet plume emitted from a spacecraft retro-propulsion rocket nozzle and the surface of a planetary body during a powered *soft landing* procedure. An impinging plume threatens mission safety and success by lofting regolith dust sheets [1] and therefore eliminating landing visibility, propelling unconsolidated surface material horizontally away from the landing site [2] which may collide with local hardware and even orbiting equipment [3], capturing regolith material in regions of recirculated flow [4] beneath the landing spacecraft and thereby causing it to be struck by *ejecta* streams, and excavating large quantities of regolith soils to form craters [5] and unstable surfaces. A further point of concern is that regolith dust species are generally highly abrasive due to the absence of any natural or aeolian erosion regime on most planetary bodies, rendering delicate equipment useless on its impact and abrading any exposed surfaces. Regolith dust is also prone to entering mechanisms and seals, resulting in the reduced operability or even premature failure of on-board mechanical devices [6]. The study of PSI is therefore of vital import to future manned and unmanned missions to land on planetary bodies, and failure to predict the behaviour of PSI and adapt spacecraft design methodologies and operational procedures accordingly will invariably result in mission failure [7].

Ultimately proven successful, the numerical simulation method, which primarily involved the use of robust commercially-available Computational Fluid Dynamics (CFD) codes, was utilised in a preliminary evaluation of a novel theorised method of mitigating negative plume-surface effects by leveraging secondary flow [8] to capture regolith *ejecta*, and simultaneously using *supersonic fluidic injector* technology [9] to minimise plume-driven surface alteration. Unique to this approach was the intended realisation of these goals using only spacecraft-borne hardware, unlike the innumerous existing proposed methods of controlling PSI which involve the use of significant surface and engineering infrastructure [10] [11]. This mitigation method was found to be remarkably effective in reducing the negative flow effects encountered during a powered soft-landing of a spacecraft and undoubtedly merits further study.

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Acronyms

CAD Computer-Aided Design.

CD Convergent-Divergent.

CFD Computational Fluid Dynamics.

HPC High-Performance Computing.

MLE Mars Landing Engine.

MOC Method of Characteristics.

MSL Mars Space Laboratory.

NS Navier-Stokes.

PAL Planetary Aeolian Laboratory.

PLIF Planar Laser-Induced Fluorescence.

PM Prandtl-Meyer.

PSI Plume Surface Interactions.

RANS Reynolds-Averaged Navier-Stokes.

SST Shear-Stress-Transport.

TCT Total Computational Time.

TRL Technology Readiness Level.

Nomenclature

β	Thermal Expansion Coefficient
ϵ	Nozzle Expansion Ratio
Γ	Specific Heat Ratio
λ	Thermal Conductivity
ϕ	Viscous Dissipation Factor
ρ	Density
τ	Viscous Shear Stress Tensor
$A*$	Nozzle Throat Area
$A_\mathrm{e}/A*$	Nozzle Expansion Ratio
A_e	Nozzle Exit Area
d	(Primary) Nozzle Exit Diameter
d_inj	Injector Nozzle Exit Diameter
h/d	Normalised Altitude
M_e	Nozzle Exit Temperature
P_e	Nozzle Exit Pressure
P_a	Ambient Pressure
P_c	(Primary) Thruster Chamber Pressure
P_g	Ground Pressure due to Impingement
T_c	Thruster Chamber Temperature
T_e	Nozzle Exit Temperature
x	Axial Length along Plume or Nozzle Axis

1 Introduction

A spacecraft retro-propulsion landing engine emits a jet of propellant gases which forms a high-enthalpy gas plume as it expands, moving at supersonic or hypersonic gas velocity. During the penultimate stage of the powered descent and soft-landing of a spacecraft, the spacecraft will have a sufficiently low altitude that its exhaust plume strikes the planetary surface in a complex and non-linear process involving multiple shock-shock and shock-surface interactions. The resulting interplay between soil physics and gas dynamics falls under the field of Plume Surface Interactions (PSI), the study of which encompasses analysing, simulating, and numerically predicting the effect of exhaust plume impingement on various surface topographies. Plume-surface effects can be divided into categories of plume flow physics, *ejecta* dynamics, and cratering [12], where the study of each typically entails the development of complex numerical models forced using recorded flight data and images. Due to accelerated initiatives to launch new crewed missions to the Moon [5] and pioneer crewed missions to Mars, the study of PSI is a field experiencing rapid growth.

The superficial layers of disintegrated surface material blanketing most terrestrial bodies are comprised of abrasive [6] species of rock- and glass-like [13] materials termed *regolith*, which range in physical size from fine-grained particles akin to dust to coarse subsurface soils and small pebbles [14]. The action of an impinging plume is to first loft and propel the dust-like surface material that constitutes the uppermost region of regolith, before beginning to erode and excavate the unconsolidated material that lies beneath it until various layers of subsurface soils are exposed. The ultimate effects are the alteration of surface topology and the excitation of regolith projectiles, described as *ejecta*. These behaviours are exhibited in Figure 1 overleaf, which depicts a hot-fire thruster test performed using Lunar regolith simulant to reproduce plume-surface effects encountered during the Apollo 12 mission [1]. The most pronounced mechanism of surface alteration is the formation of craters, where plume-driven ejecta presents a dangerous condition to the landing spacecraft. This is a consequence of crater geometry recirculating and redirecting plume flow towards the spacecraft, causing it to be struck by gas *streams* containing abrasive dust and debris moving at high velocity. A similar threat is posed to structures and equipment situated on the surface local to the landing site, as plume flow is turned horizontal at the point of impingement and propagates radially outward, carrying dust and other small particles at an often supersonic velocity. For a planetary body of sufficiently small mass, plume-driven regolith dust may be propelled at or above planetary escape velocity, extending this hazard to orbiting hardware [4]. Freely evident is the

danger inherent to plume-surface effects and the reason for its exhaustive study by innumerous space agencies. It is also readily understood that the ability to reliably predict the surface-altering behaviour of plume impingement is imperative for the success of future missions [7].

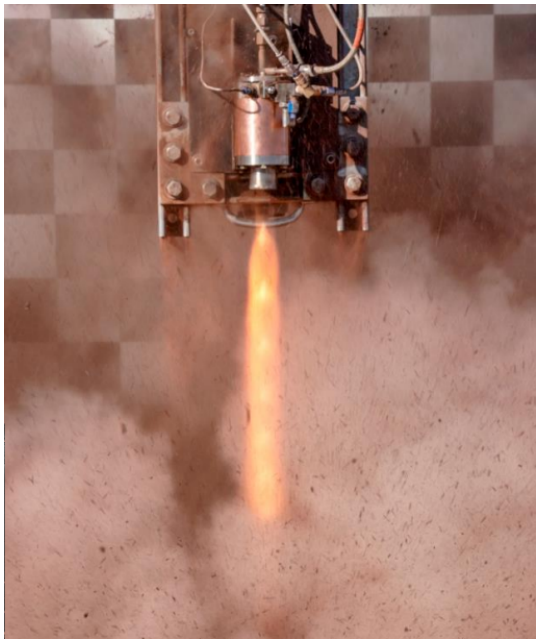


Figure 1: Reproduced Landing Engine Plume Impingement to Represent *Apollo 12* Mission using Lunar Regolith Simulant. Figure Prepared by Kuhns *et al.* (Available in Reference [1])

The primary focii of the global study of PSI, the Moon and Mars, will be considered in this work. The specific behaviours of PSI are dictated by the atmospheric environment individual to a planetary body and the soil properties at its surface, as well as the dry mass of the lander [15]. For instance, the presence of a thin atmosphere on Mars serves to *collimate* (or concentrate and align parallel) the gas plume, resulting in stronger shock wave interactions and greater impingement pressures when compared to the equivalent scenario for Lunar vacuum conditions [16]. An alternative factor is that Martian regolith has greater porosity and is less cohesive than Lunar regolith, causing it to be more susceptible to excavation by an impinging plume [4]. The study of plume-surface effects under Martian conditions therefore yields greater-fidelity results versus Lunar conditions by virtue of more easily discernible gas flow phenomena and soil movement regimes, hence the focus of this study will be chiefly, but not solely, directed toward spacecraft landings on Mars.

The majority of proposed strategies to combat the negative effects of PSI range from the construction of artificial landing surfaces using erosion-resistant materials [4] to the forming of boundary

walls to shield locally incident equipment and structures from horizontally-propagating plume flow [11]. A commonality shared between these concepts is the existence of significant operational difficulties stemming from the requirement to develop engineering devices and transport them to the Moon or to Mars to implement each strategy, wherein these missions would similarly encounter destructive plume-surface effects. Of the very few spacecraft-based mitigation strategies recommended, all were of a low Technology Readiness Level (TRL) [4].

1.1 Project Aims and Objectives

- The first aim of this study is to create and validate a numerical method of simulating impinging hypersonic plume flows.
- The second aim is to use this method to develop and undertake a preliminary evaluation of a mechanism by which the destructive actions of PSI may be controlled via manipulation of the exhaust plume using only spacecraft-borne equipment, reducing the need for surface-based infrastructure. It is hoped that the solution will be realisable by adapting the existing function of a retro-propulsion landing engine nozzle. Of note is the improbability that any one strategy will solve the problem of PSI, where a more realistic approach would be to implement both spacecraft-based and minor surface-based mitigation methods simultaneously. It is theorised that by deliberately introducing a strong swirl velocity to the plume flow at the landing engine nozzle exit, the viscous shear forces that are assumed to exist between the rotating and expanding gas plume and the quiescent atmospheric gases of a planetary body will induce a smaller angular velocity at the perimeter of the plume than at its core. From a consideration of essential fluid dynamics, it is understood that a small gas velocity corresponds to large gas pressure. Thus, the velocity differential between the plume core and perimeter, a coupled rotating structure that can be described as a *vortex*, should result in the existence of a pressure force acting radially inward towards the centre of the plume. This pressure field is expected to contain, or at a minimum decelerate, the aforementioned regolith ejecta, preventing it from striking both the landing spacecraft and nearby hardware. Furthermore, for the situation of PSI on Mars, introducing a rotational component to the gas velocity is anticipated to disrupt the collimation of the impinging plume, and thus act to reduce the severity of the induced cratering by reducing the total pressure incident at the point of impingement. The gas swirl velocity may be induced by siphoning a controlled amount of gas flow from the nozzle plenum and transporting it just beyond the nozzle exit plane, directing it into the exhausted gas plume at a set angle of incidence.

2 Literature Survey

The following sections present a concentrated summary of the literature reviewed prior to beginning this study, divided into common areas to introduce in chronological order the topics to be discussed later in this work.

2.1 Contemporary Knowledge of PSI Phenomenology

The information reproduced in the following sections is intended to: 1) relay documented plume-surface effects and behaviours, 2) develop a working understanding of the physics driving PSI, 3) provide an overview of the operational challenges encountered due to plume-surface effects, 4) discuss existing methods of mitigating plume-surface effects and 5) present a novel solution to PSI.

2.1.1 Observed Plume-Surface Effects on the Moon

The lack of a substantive atmosphere on the Moon results in a highly expanded exhaust plume of rarefied gas [3] which impinges across a large surface area. Lunar regolith is cohesive and dense, with a layer of extremely fine, highly-abrasive [6] dust-like material blanketing the surface. Particles of this species of *ultra-fine* meteoric impact-produced glass are generally below 10 μm [13] in radius. Plume-surface effects are first observed when the spacecraft descends to an altitude of 30-40 m [4] above the Lunar surface, wherein impinging plume flow lofts and propels erratically the ultra-fine dust particles, diminishing visibility during landing. At lower spacecraft altitudes, radially-propagating redirected plume flow has sufficient energy to propel dust-to-sand-sized *fine* particles (typically less than 20 μm in radius [14]) across the Lunar surface, at velocities reaching 3 km s^{-1}. For a strong plume flow corresponding to a large spacecraft landing mass, the <20 μm fraction of Lunar dust, which comprises the upper few centimetres of regolith [4] or approximately 20% of Lunar soils by weight [13], can be propelled at or above Lunar escape velocity and become inserted into a low orbit [2].

As the spacecraft descends further and impingement pressures increase, the plume transports away the fine and ultra-fine particles on the surface, and comparatively coarse subsurface soils (of greater than 50 μm in particle radius) interspersed with small rocks and mineral deposits [14] are exposed. Although the mechanism of plume-induced soil erosion does occur, the relative impermeability of Lunar regolith causes the resulting craters to be small in diameter and depth. However, the effect of cratering is still magnified with increasing lander dry mass [15]. The geometry of a crater formed

below a spacecraft results in gas jets containing regolith material being redirected towards the spacecraft [4]. Plume flow propagating horizontally away from the landing site has been observed to contain significant amounts of regolith material and small rocks, where Metzger *et al.* noted that the exhaust plumes of the Apollo 12 Lunar Module propelled rocks as large as 10 centimetres in diameter across the Lunar surface [2].

2.1.2 Dissimilarities of Plume-Surface Effects on Mars

The presence of an atmosphere on Mars, albeit thin, serves to worsen the effects of PSI compared to the same behaviours observed on the Moon by way of *collimating* (or aligning parallel to itself) the plume flow, though the plume-surface effects induced are superficially similar.

Collimation of the exhaust plume results in a denser plume than is observed for Lunar conditions, where focused impingement pressures yield greater gas total pressure exerted on the impingement surface and stronger shock wave interactions [16] [4]. Furthermore, Martian regolith is less cohesive than Lunar regolith and possesses greater gas permeability [4], and as such is more susceptible to excavation and transportation by an impinging jet. Consequently, plume impingement under Mars ambient conditions gives rise to intensified material excavation with comparatively increased rates of erosion [1] for a given surface pressure and spacecraft landing mass, with broadly exacerbated plume-driven topographic alteration.

Conjointly, exhaust plume collimation results in the initiation of PSI at greater altitudes for Martian atmospheric conditions than for Lunar conditions: plume-surface effects on Mars have been observed at altitudes as high as approximately 63 metres [12]. The first observed plume-surface effect at this altitude is the lofting and excitation of *ultra-fine* dust-like material, in a condition related to the lofting of Lunar regolith dust sheets that rapidly diminishes visibility. It is of note that particles of Martian dust and soil species are generally smaller than their Lunar counterparts, with the *ultra-fine* Martian dust species consisting of particles of the order of 2 μm in radius [17] versus the approximately 5 μm in particle radius for Lunar dust. Moreover, the presence of a thin atmosphere on Mars will invariably have subjected Martian regolith to a weak aeolian erosion regime, therefore Martian regolith is accordingly less abrasive than Lunar dust but can still abrade sensitive spacecraft hardware. Furthermore, unlike the Lunar vacuum which allows disturbed particles to move unimpeded, the presence of Mars' atmosphere causes lofted Martian regolith to decelerate rapidly, and the mass of Mars results in an escape velocity unattainable by

the *fine* species of Martian dust that tend to be greater than 5 μm in radius [17]; these previous two conditions lend themselves to diminishing the interaction mode of Martian dust-lofting.

At spacecraft altitudes closer to the Martian surface, the impinging plume excavates and propels unconsolidated surface material, and rapidly erodes the subsurface soils that lie beneath to form large craters with material deposited around the crater rim [5]. Although analogous to Lunar plume-surface interactions, these phenomena proliferate in a shorter period of time on Mars than they do on the Moon. Equivalently, high-velocity ejecta streams containing large quantities of regolith material arc towards the landing spacecraft, directed by crater geometry. Martian regolith is less dense than Lunar regolith, wherein ejecta streams on Mars carry accordingly greater quantities of regolith than they do on the Moon which strikes the under-body of the landing vehicle.

2.1.3 Operational Consequences of PSI

Plume-surface effects experienced for Lunar and Martian environments are largely identical, hence their impact on spacecraft operation will be described in tandem to minimise repetition.

The lofting of regolith dust sheets introduces a dangerous condition for optically-guided spacecraft landings, as the fine particles completely obscure the surface for the duration of the soft-landing procedure due to excitation by the impinging jet [1]. These dust sheets contain irregularly-shaped, sharp-edged particles which strike and abrade exposed surfaces, and are particularly harmful to imaging lenses and other optical devices [4] with polished surfaces. A similar consideration is that regolith dust is prone to enter mechanisms, hinges, bearings, and seals, which often results in the reduced operability or even premature failure of on-board mechanical hardware and manipulators [6]. It is prudent to reiterate that the abrasive effect of lofted dust is heightened for Lunar regolith due to its composition of jagged-edged glass particles that have not experienced any form of natural erosion [6]. When propelled by strong plume flow on the Moon or another terrestrial body of similarly small mass, this material can become inserted into Lunar orbit and can therefore strike orbiting hardware, threatening significant damage [2].

At altitudes closer to the planetary surface, impinging plume flow propagates horizontally away from the landing site, travelling along the surface and propelling loose material [2]. For Lunar PSI, the presence of the Lunar vacuum enables plume-driven ejecta to travel unimpeded, extending the threat of ejecta impact to ground-based hardware situated within a few hundred metres of the landing site [18]. Indeed at this distance, impact from plume-driven regolith has an effect

comparable to a coarse media abrasion process, albeit increased in intensity by several orders of magnitude. This is unambiguous in Figure 2 overleaf, which depicts a panel of the NASA *Surveyor III* Lunar lander which sustained damage from plume flow containing regolith material, which propagated from the NASA *Apollo 12* mission landing site located approximately 180 m away [19]. Evident in this image is the surface deterioration due to plume spray, where regions of compromised surface are highlighted and contrasted against an area of undamaged surface.

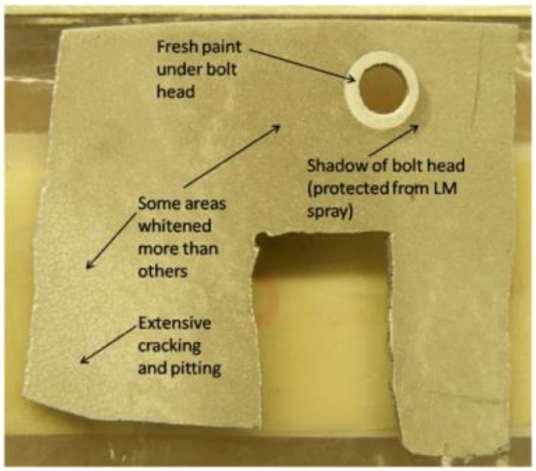
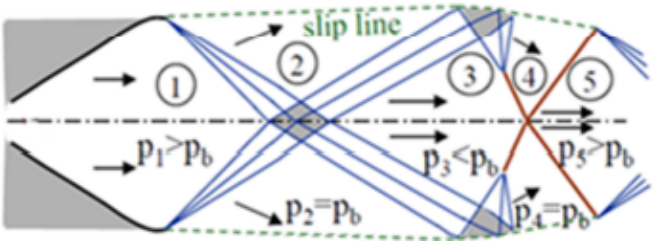


Figure 2: Operational Impact of Lunar PSI: Damage to Surveyor III. Figure Prepared by Watkins *et al.*, 2021 (Available in Reference [4]).

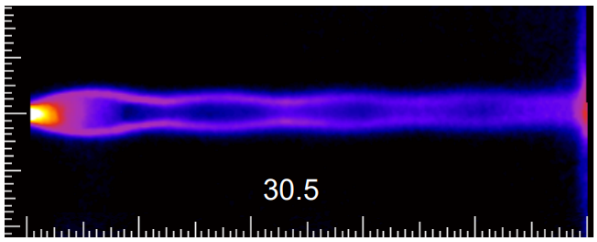
The formation of craters by plume-induced erosion and regolith transportation introduces surface geometry that precipitates the redirection of plume spray towards the landing spacecraft [4] as discussed previously, containing large amounts of entrained regolith material. Collisions from high-velocity ejecta wear and destroy equipment mounted on-board the landing spacecraft. Under-slung hardware is especially vulnerable, demonstrated when one of the two anemometers on-board the *Curiosity* Mars rover became permanently disabled following underbody ejecta impact [20] during its soft-landing by the *SkyCrane* vehicle. A further consideration is that the excavation and relocation of surface material immediately beneath a landing spacecraft can result in an unstable, uneven surface on which the spacecraft will rest, potentially jeopardising a mission by placing under-slung propulsive or scientific equipment undesirably close to surfaces. This scenario resulted in the structural failure of the NASA *Apollo 11* Mission Lunar Excursion Module (LEM) descent engine nozzle skirt [12], wherein its proximity to the crater rim caused the descent engine plume flow to back-fill and over-pressurise the nozzle. An unstable or unpredictable landing surface may also result in a spacecraft inclination sufficiently extreme to render optical equipment useless [1].

2.1.4 Physics and Flow Structure of Supersonic Plumes

Landing engine exhaust jets are under-expanded [21], which will be the sole type of flow detailed here in the near-field regime. For an under-expanded jet, which is characterised by a cyclic arrangement of gas compression and expansion flow structures, the exit jet pressure is, by definition, greater than ambient pressure. This leads to the formation of Prandtl-Meyer (PM) several expansion waves initiated at the nozzle lip [16], referred to collectively as an *expansion fan*, which acts to reduce the pressure in the gas jet to ambient pressure. This occurs at Location 1, illustrated in Figure 3a overleaf. As can be seen in Figure 3a, this expansion fan turns the flow away from itself as the flow is accelerated, growing the jet boundary [22].



(a) Flow Structure Schematic of an Under-Expanded Supersonic Jet; Figure Prepared by Inman *et al.* (Available in Reference [23]).



(b) Planar Laser-Induced Fluorescence (PLIF) Image of the Flow Structure of an Under-Expanded Supersonic Jet; Figure Prepared by Inman *et al.* (Available in Reference [23]).

Figure 3: Structure of the Gas Jet Plume of an Under-Expanded Supersonic Jet.

The reflection of PM expansion waves in Location 2 leads to a gas jet pressure less than ambient pressure, denoted here as P_b, at Location 3. Consequently, expansion waves from the ambient pressure boundary are reflected [16] and form Mach waves (or compression waves), which increase the gas pressure at Location 3, marked P_3. These Mach waves consolidate to form strong, oblique shock waves, which act to match the pressures incident at Locations 3 and 4 (P_3 and P_4 respectively), however, the shock-shock interaction at Location 4 results in a gas pressure greater than ambient pressure at Location 5. It is intuitively understood that as $P_5 > P_b$, this process will renew in a series of train-like repeating flow structures referred to as *Shock cell* trains [16] [23] as P_5 converges to P_b. Shock cells dissipate downstream due to viscous losses and turbulent mixing with the atmosphere. Depicted in Figure 3b is a Planar Laser-Induced Fluorescence (PLIF) image of a real shock-cell train for reference, shown at a horizontal distance of 30.5 nozzle exit diameters, d, from the nozzle exit plane to the impingement plate. An additional flow structure termed a *Mach disc* is often formed for high nozzle pressure ratios [22], which is the coalescence of reflected

Mach waves propagating from the curved boundary of the jet shear layer. The convergence of these Mach waves is presented as a strong *standing shock* oriented perpendicular to the direction of flow, depicted in Figure 4 below.

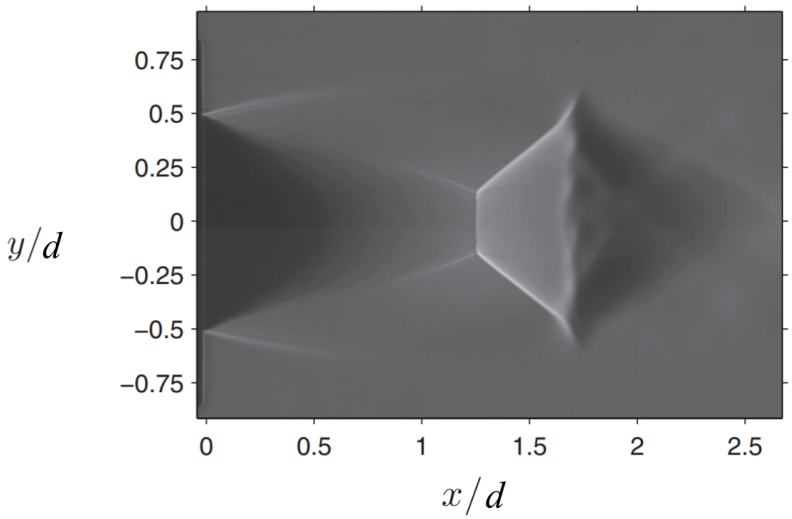
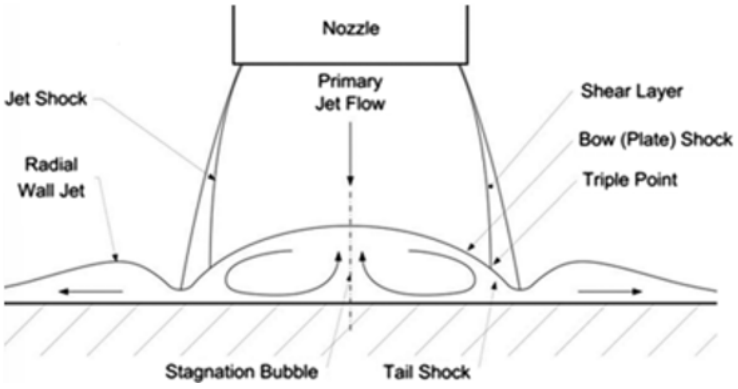


Figure 4: Schlieren Image of Density Change in the Direction of Flow, $\partial\rho/\partial x$, Shown for Mean Flow. Figure Adapted from Reference [22].

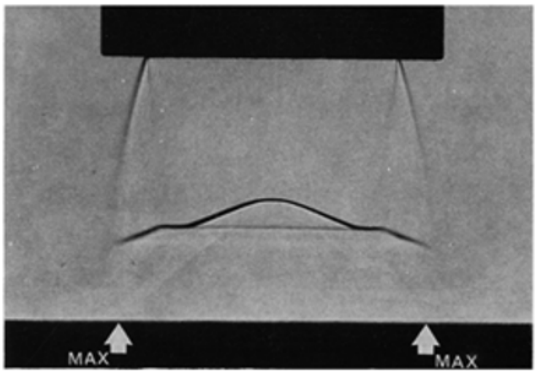
Figure 4 presents a Schlieren image of an under-expanded flow with a Mach disc located at a normalised distance of $x/d \approx 1.4$, where d represents the normalising parameter of nozzle exit diameter. Note that the plotting of density change in a flow field is done to reveal shock structures that are not usually obvious. It can be seen that plume flow immediately downstream of the Mach disc is substantially decelerated, although the mean flow velocity recovers due to viscous mixing further downstream. Strong oblique shock waves are present above and below the disc, intersecting at the *triple point* from which the annular shear layer of the disc propagates [22], with a series of reflected and coalescing shock waves downstream. A triple point in supersonic flows is the point of intersection of an incident and reflected shock wave [24], as well as the perimeter of a *slipstream* separating flow regions of disparate velocity.

2.1.5 Physics of Jet Impingement

Shown overleaf in Figures 5 is the structure of a descent engine exhaust plume in the far-field region when it impinges on a surface. Figure 5a depicts a schematic prepared by Pattison *et al.* [25] of the flow structure, and Figure 5b depicts an image of impinging flow captured using the Schlieren imaging technique, prepared by Lamont and Hunt [26]. Indicated in Figure 5a is



(a) Flow Structure Schematic of an Impinging Jet Prepared by Pattison *et al.* (Available in Reference [25]).



(b) Schlieren Image of Flow Structure Prepared by Lamont and Hunt (Available in Reference [26]).

Figure 5: Flow Structure of Impinging Supersonic Plume Flow, Far Field

a tail shock wave, a variety of oblique shock wave that stems from the *triple point*. Here, the triple point is the point of intersection of the bow shock, the shear layer between the supersonic plume flow and surrounding atmosphere (or *slipstream*), and the tail shock wave. The bow shock introduces a surface of high gas pressure below which the flow cannot overcome the resulting strong pressure gradient, causing the flow below the bow shock to be recirculated [12] [16]. This area of flow is referred to interchangeably as the *recirculation zone* or *stagnation bubble*, and is a critical component of plume-driven soil erosion due to the large surface pressures applied by the associated shock-shock and shock-surface interactions. The Schlieren image of the same flow structure shown in Figure 5b contains clear outlines of the shock waves and corresponding shear layers. Of particular prominence are the radial wall jets and bow shock, illustrating the mechanism by which impinging exhaust plumes apply destructive surface pressure and simultaneously loft and propel regolith dust and other small particles horizontally across a planetary surface.

2.2 Mitigating and Controlling PSI

2.2.1 Existing Proposals

Of the theorised methods of managing plume-surface effects during landing, it is often the consensus that the most optimal approach is the deployment of erosion-resistant landing pads on which spacecraft can safely and repeatedly land. This favour is the result of landing surface methods possessing the greatest Technology Readiness Level (TRL) of all proposed mitigation methods [4]. Such methods typically involve the construction of landing pads from locally available or readily

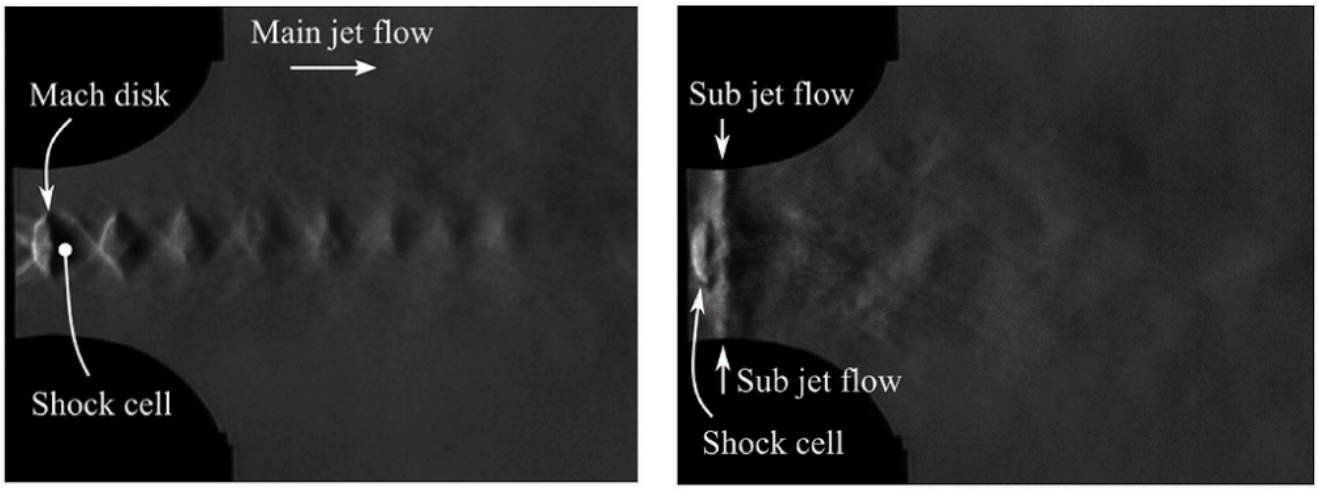
manufacturable materials, such as regolith bricks or *Martian concrete* [10]. Notable among the more advanced landing surface construction methods that have been proposed is *regolith sintering* [27] which involves harnessing solar energy to thermally bond surface regolith material, an approach which would require the least intensive civil engineering infrastructure of the theorised landing pad construction techniques. Each strategy presented here requires action to be taken only on the surface of a planetary body, and therefore all are consigned to the category of *surface-based mitigation techniques*. None of the aforementioned proposals, however, can be considered a complete solution, as all present a lack of protection for locally-incident hardware. A supplementary surface-based mitigation technique that may overcome this obstacle is the newly-suggested formation of rudimentary barrier walls [11] around landing sites, intended to shield structures from redirected plume flow. Unfortunately, common to all surface-based methods of controlling PSI is a significant operational challenge yet to be solved: each approach requires the development of substantial ground-based civil engineering infrastructure to facilitate the operation of engineering devices to implement proposed strategies. Such devices, which have yet to be developed, would likely need to operate autonomously, and would themselves have to be safely transported to the Moon or to Mars without becoming damaged or destroyed by the inevitable plume-surface effects they would encounter. A further concern is that precursor expeditions and missions may be required to prepare for any surface-based mitigation technique.

The implementation of *spacecraft-based mitigation techniques* would eliminate the infrastructure-dominated operational challenges inherent to surface-based mitigation techniques, although few such methods have been theorised and developed. The leading proposal for a spacecraft-based mitigation technique is the injection of engineered alumina particles into thruster plumes [4], whereby metallic compounds are introduced to the plume flow which coat and bond with surface materials immediately below the spacecraft, transforming regolith materials into a type of instantaneous landing surface. Although expected to be highly effective due to a greatly improved thermal and ablation resistance [28] com, this is at the extreme cost of polluting a planetary surface and rendering local scientific surface studies impossible.

2.2.2 Supersonic Fluidic Injectors

A contemporary study by Kobayashi and Ukai entitled "Effectiveness of a Supersonic Jet-Jet Interaction Technique on Flow Structure in an Underexpanded Jet Impinging on a Perpendicular Plate" explored a promising technology relevant to spacecraft-based mitigation techniques. The

work of Kobayashi and Ukai described the use of *supersonic fluidic injectors* in reducing transient impingement pressures and minimising shock cell structures, available in Reference [9]. It was found that equipping multiple supersonic fluidic injectors firing perpendicular to the flow produced an advantageous jet-jet interaction, where the flow became increasingly radially expanded with no shock-cell structures beyond the line-of-action of the injectors, and thus no standing shock (or bow shock) was formed above the recirculation zone on the impingement plate. This flow behaviour is exhibited in Figures 6a and 6b below, which exhibit the same flow without and with the action of supersonic fluidic injectors respectively. Apparent in Figure 6a is the shock cell train, notably absent in Figure 6b, which also illuminates the approximate location and function of the injectors.



(a) Free Jet Expansion: without Jet-Jet Interaction. (b) Free Jet Expansion: with Jet-Jet Interaction.

Figure 6: Instantaneous Schlieren Image of Expanding Free Jet: Influence of Supersonic Fluidic Injectors. Figures Prepared by Kobayashi and Ukai (Available in Reference [9]).

The use of supersonic fluidic injectors was observed to decrease the maximum stagnation pressure at the impingement point by 48%, a factor invaluable in minimising plume-surface effects as plume-induced cratering is driven by large surface pressures [12]. This reduction in pressure was at the negligible cost of reducing the descent engine thrust by 6.1%. In the case of Kobayashi and Ukai's study, gas flow to the supersonic injectors was prescribed for the same chamber conditions as the primary nozzle, a scenario which lends itself well to siphoning flow from the nozzle plenum to implement this technology in practice. This is readily understood from Figure 7, available overleaf, which depicts the arrangement of apparatus employed for the work and includes the positions of the fluidic injectors relative to the primary nozzle.

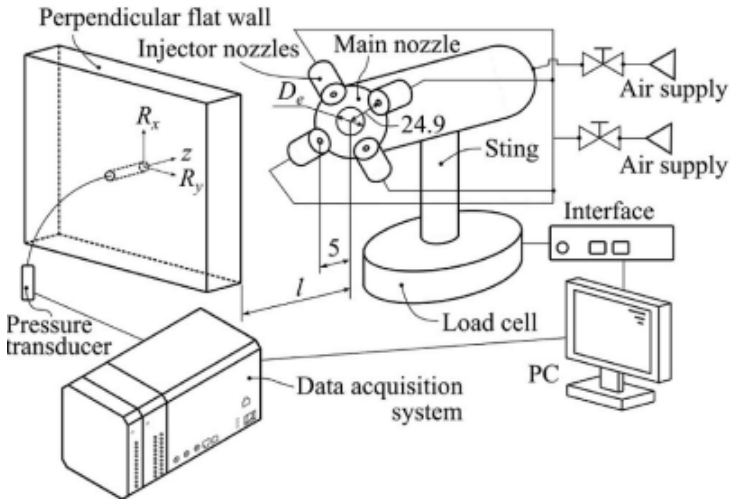


Figure 7: Experimental Schematic of Supersonic Fluidic Injector Study. Figure Prepared by Kobayashi and Ukai (Available in Reference [9]).

2.2.3 Applied Literature: Canted Supersonic Fluidic Injectors

As discussed in Section 1, the aim of this work is to theorise and evaluate a novel method of manipulating plume flow to reduce the severity of plume-surface effects during landing. This may be achieved by adapting the supersonic injector technology analysed by Kobayashi and Ukai to leverage secondary flow to induce an ejecta-containing pressure field.

Reviewing Figure 7 above, it is theorised that by rotating the injector nozzles about the z-axis defined in the Figure, a rotational component of the plume gas velocity will be introduced, referred to in an aerospace context as a *swirl velocity*. It is understood from Einstein's 1926 work on the effects of *secondary flow* (available in Reference [8]) that viscous effects between a rotating fluid, a structure termed a *vortex*, and a static wall induce a lower velocity at the perimeter of the vortex than at its core, resulting in a pressure force acting radially inward. The intended arrangement of forces and velocities is illustrated in Figure 8 overleaf. Although in the case of mitigating plume-surface effects the static wall is substituted for a static atmosphere, viscous effects must exist between the high-enthalpy plume flow moving at supersonic velocity and the quiescent atmosphere; this may result in a weaker pressure field than that which occurs for a rotating fluid enclosed by walls, but it is hoped that the pressure field will have sufficient strength to decelerate, if not contain, plume-driven ejecta. The existing influences of fluidic injectors on the supersonic plume are assumed to act in this scenario too, yielding a reduced shock wave-induced ground pressure (thus reducing plume-driven surface alteration) to complement the artificial pressure field.

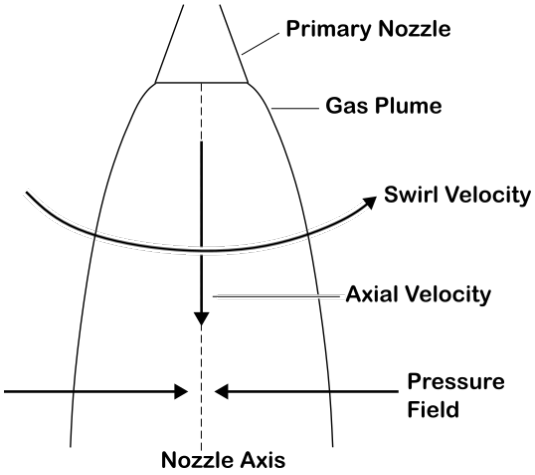


Figure 8: Arrangement of Swirl and Axial Velocities including Pressure Field, for Canted Supersonic Fluidic Injector Technology

The reduction of impingement surface pressure is expected to reduce the effect of plume-driven cratering, in turn minimising the redirection of plume flow to negate plume spray towards the landing spacecraft. In conjunction, the induced pressure field is expected to contain ejecta driven by radially-propagating redirected plume flow, reducing the risk of regolith impact posed to hardware close to the landing site.

2.3 Literature for the Development of Numerical Methodologies

To aid in the development and validation of a method of numerically simulating nozzle flow, "Design of a Supersonic Nozzle" by A. McCabe, available in Reference [29], was analysed. Similarly, to expand this method to include impinging flows relevant to spacecraft retropropulsion, "Thruster Plume Surface Interactions: Applications for Spacecraft Landings on Planetary Bodies" by M. Mehta *et al.*, available in Reference [16], was studied.

2.3.1 "Design of a Supersonic Nozzle" by McCabe, 1967

A supersonic nozzle was designed using then-leading-edge rigorous numerical methods by A. McCabe in 1967. His work is available in Reference [29]. The nozzle geometry was obtained using the full Method of Characteristics (MOC) approach, adapted to incorporate updated compressible flow theories where continuous nozzle curvature was guaranteed by specifying a continuous Prandtl-Meyer (PM) angle gradient along the nozzle axis. The result of McCabe's work was the nozzle geometry for a minimum length, boundary-layer-corrected supersonic nozzle geometry for Mach 5

nozzle exit flow. McCabe's exhaustive numerical design methodology ensured the development of an exceptionally accurate nozzle operating at near-optimal parameters across its length, a desirable property for an initial attempt at simulating nozzle flow. The feature that made McCabe's work invaluable to this study was the availability of x-y data corresponding to the nozzle profile, which are not often made available. Conveniently, the design nozzle exit velocity prescribed is complementary to aerospace applications, though the intention of the McCabe nozzle was for use in supersonic wind tunnel testing.

Available below in Figure 9 is a schematic representative of McCabe's final nozzle geometry, including *characteristics* and the nozzle streamline. Note that the nozzle curvature depicted in Figure 9 does not reflect the exact geometry published in the work, and instead serves to illustrate the design methodology employed by McCabe. For completeness and reference, the published 2-D coordinate data are reproduced in Appendix A. Assessment of the validity of a numerical simulation method developed using McCabe's work will consist of comparing the obtained nozzle flow against the design flow properties published in the work, such as the exit Mach number, as well as against existing literature regarding supersonic nozzles. For instance, it is understood that physical nozzle flow should exhibit a uniform acceleration across its length [30], and according to testing carried out by McCabe, flow at the nozzle exit of this geometry should be uniform across its diameter [29].

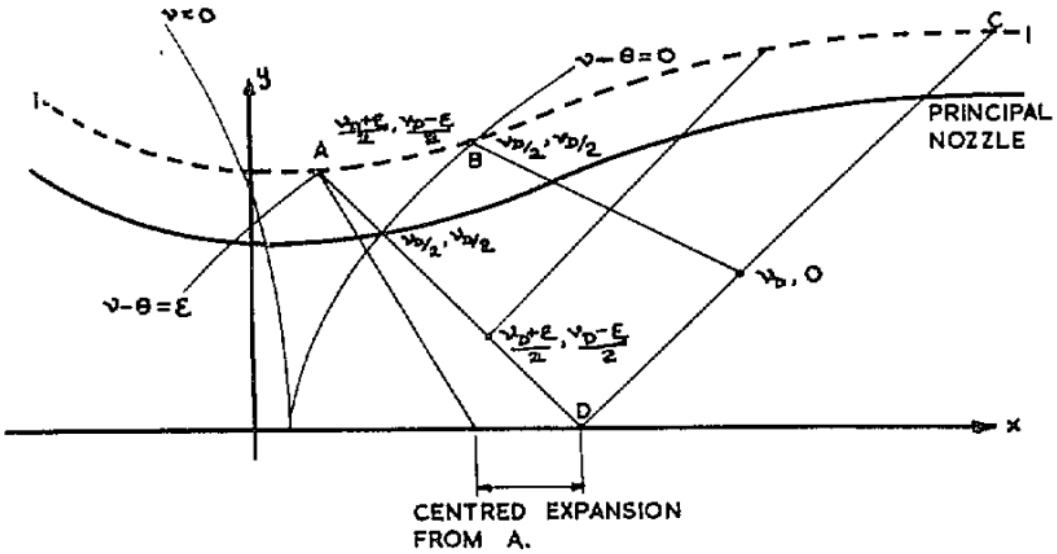
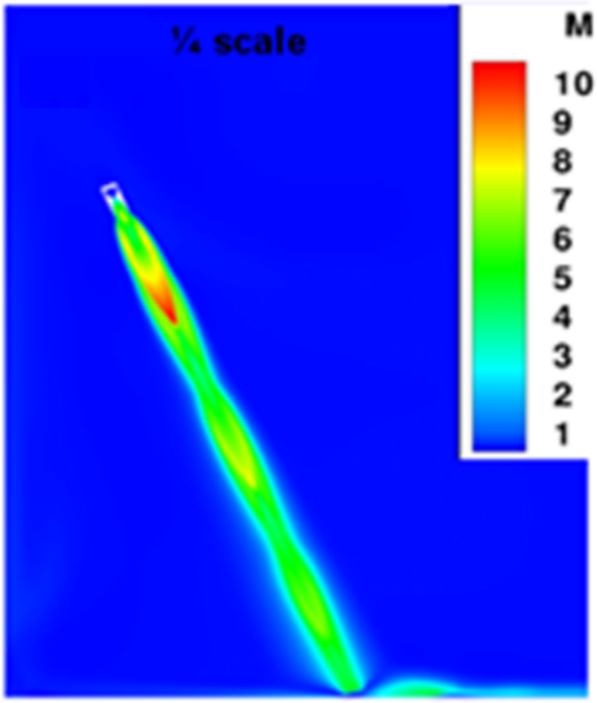


Figure 9: Supersonic Convergent-Divergent Nozzle Geometry Developed by A. McCabe, 1967. *Characteristics* for MOC design approach detailed. Figure Prepared by McCabe (Available in Reference [29]).

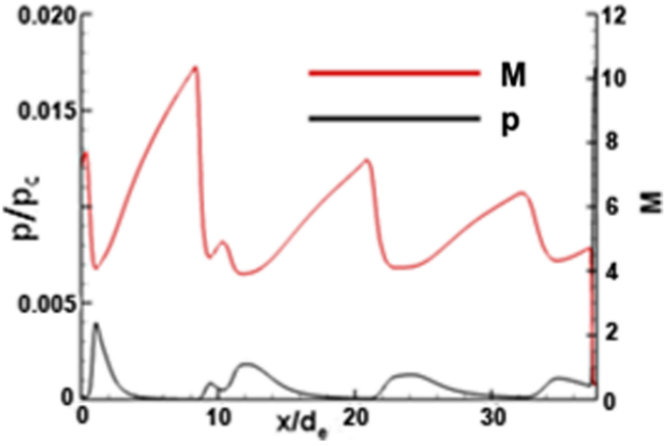
2.3.2 "Thruster Plume Surface Interactions: Applications for Spacecraft Landings on Planetary Bodies" by Mehta *et al.*, 2013

An investigation was carried out by Mehta *et al.* with the goal of examining supersonic jet interactions with a flat plane for various atmospheric conditions and landing engine configurations, available in Reference [16]. Of particular interest was the work done to numerically simulate the exhaust plume impingement of one of the NASA Mars Space Laboratory (MSL) spacecraft's eight Mars Landing Engine (MLE)s during the final deceleration phase. This craft is better known as the NASA *SkyCrane* descent vehicle used to execute the soft-landing of the NASA Mars rover *Curiosity*. To reproduce the conditions encountered by the *SkyCrane* vehicle, relevant numerical simulations were computed using Mars surface conditions with simulation parameters informed by recorded flight data from previous Mars missions. The results of these simulations were compared against experimental data obtained at the NASA Planetary Aeolian Laboratory (PAL) vacuum chamber facility for a quarter-scale model of the MSL MLE nozzle. The experimental nozzle was oriented to match the 22.5° angle of inclination of the *SkyCrane* landing engine nozzles, and fired at a normalised altitude of 34.5 nozzle exit diameters, d, above the impingement plate; this altitude was prescribed to replicate the lowest altitude at which the *SkyCrane* vehicle hovered whilst landing *Curiosity*, effectuating a worst-case scenario. The numerical simulation was synthesised using identical nozzle geometry and general arrangement, and both the numerical simulation and experimental model incorporated nozzle chamber properties to satisfy 100% throttle conditions, prescribed to represent the true soft-landing procedure. Conveniently, numerical results were produced for quarter-scale nozzle geometry to match the quarter-scale experimental results.

Ideal for the purpose of developing and validating a computational method of numerically simulating flows in the far-field regime, encompassing plume flow and impingement flow, were the simulated exhaust plume Mach contours produced for the sub-scale MSL MLE nozzle by Mehta *et al.* This was due to the abundance of supplied nozzle and simulation parameters and the availability of experimental hot-fire test data for direct comparison. Presented in Figure 10a overleaf are the plume gas Mach contours for the impinging MSL MLE plume using sub-scale nozzle geometry. Also included overleaf in Figure 10b is a plot of gas Mach number and gas pressure, normalised against chamber pressure, P_c, against horizontal distance from the nozzle exit plane to the impingement plate. Although produced in 3-D by Mehta *et al.*, careful selection of spatial and physics models would enable the same simulation to be evaluated in 2-D space, facilitated by the



(a) Quarter-Scale MSL MLE Numerical Simulation: Exhaust Plume Mach Contours. Figure Courtesty of Mehta *et al.*, 2013 (Available in Reference [16]).



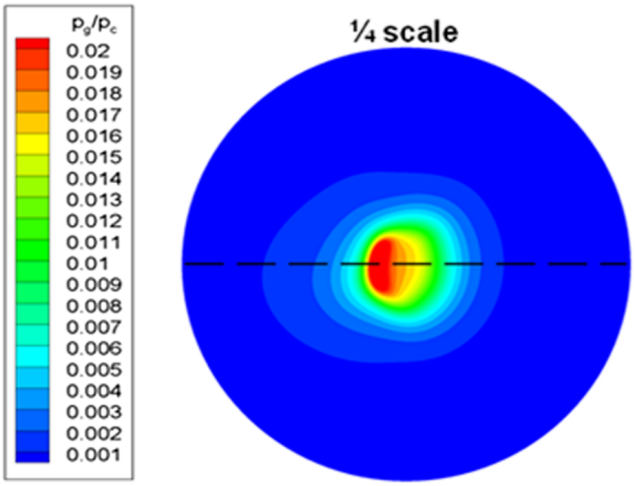
(b) Quarter-Scale MSL MLE Numerical Simulation: Plume Gas Mach No. and Gas Pressure versus Distance from Nozzle Exit Plane. Figure Courtesty of Mehta *et al.*, 2013 (Available in Reference [16]).

Figure 10: Simulation Results for Quarter-Scale Mars Space Laboratory (MSL) Mars Landing Engine (MLE) Nozzle Geometry Obtained by Mehta *et al.*, 2013: Exhaust Plume

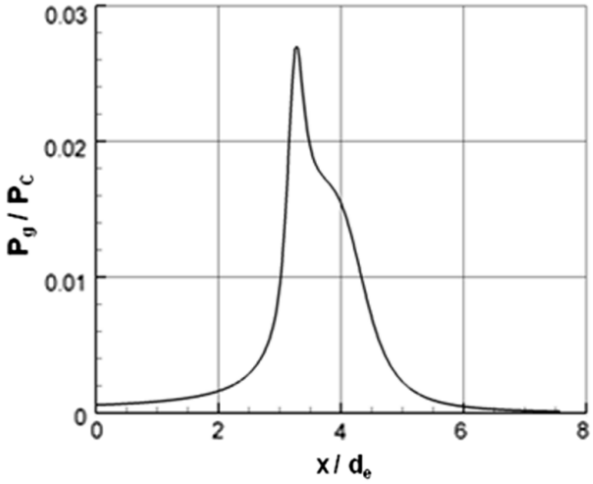
presentation of the published results in the x-y plane. For the purposes of generating numerical simulation methodology, reproducing the above results in 2-D would allow for a direct expansion of the simulation method to be developed to replicate near-field flow through the McCabe nozzle (see Section 2.3.1) to then be applicable to far-field flow regimes. The built-in validation of employing the results obtained by Mehta *et al.* to develop a numerical simulation method will arise from equipping the method to reproduce the same results.

Mehta *et al.* also obtained numerical and experimental data for the plume impingement pressure distribution, provided overleaf in Figures 11a and 11b. Figure 11a depicts the profile (or *footprint*) of impingement pressure distribution, whilst Figure 11b provides a plot of ground impingement pressure, P_g, normalised against chamber pressure, P_c, versus radial distance from the plume centre-line. These data can only be replicated by considering plume flow in 3-D space and therefore will be invaluable in further expanding the to-be-developed 2-D far-field flow numerical simulation methodology to include 3-D spatial modelling. As before, the inherent validation of this

procedure will grow from using the complete method to replicate the very results that informed its development.



(a) Quarter-Scale MSL MLE Numerical Simulation: Impingement Pressure Distribution, Footprint. Figure Courtesty of Mehta *et al.*, 2013 (Available in Reference [16]).



(b) Quarter-Scale MSL MLE Numerical Simulation: Impingement Pressure Distribution, Plot Normalised against Chamber Pressure P_c. Figure Courtesty of Mehta *et al.*, 2013 (Available in Reference [16]).

Figure 11: Simulation Results for Quarter-Scale Mars Space Laboratory (MSL) Mars Landing Engine (MLE) Nozzle Geometry Obtained by Mehta *et al.*, 2013: Impingement Pressure

The full, compressible Reynolds-Averaged Navier-Stokes (RANS) equations were solved over the entire simulation domain, specified as 2.74 x 2.74 x 2.49 m (length x width x height) or 55.28 x 55.28 x 50.23 *d*. A computational mesh was built using a series of structured grids to specify geometries for the nozzle interior, nozzle enclosure, and jet plume, whose dimensions were not provided. A 3-D flow solver with a time-marching implicit scheme was prescribed, in conjunction with Menter's Shear-Stress-Transport (SST) turbulence model including compressibility correction. Conservation of total system mass was assumed, where the mass flux is balanced between domain fluid inlets and outlets. Viscous spacing and infinite *wall y+* treatments were specified at *wall* boundaries, with a total-pressure, total-temperature inflow boundary condition applied to the nozzle chamber to satisfy flow properties at the nozzle plenum.

3 Numerical Methodology

3.1 Computational Fluid Dynamics

Numerical simulations in this work were produced using *Simcenter STAR-CCM+*, a robust Computational Fluid Dynamics (CFD) suite. The governing compressible Navier-Stokes (NS) equations solved by CFD tools on average across the entire simulation domain are presented below in condensed symbolic notation (adapted from Reference [31]):

Continuity Equation :

$$\frac{D\rho}{Dt} = -\rho \bigtriangledown \cdot \overrightarrow{v} \tag{1}$$

Momentum Equation:

$$\rho \frac{D\overrightarrow{v}}{Dt} = \overrightarrow{f} - \bigtriangledown p + \bigtriangledown \cdot \boldsymbol{\tau} \tag{2}$$

employing the *viscous stress tensor*, $\boldsymbol{\tau}$:

$$\boldsymbol{\tau} = \mu(2\dot{\varepsilon} - \frac{2}{3}\boldsymbol{\delta} \bigtriangledown \cdot \overrightarrow{v}) \tag{3}$$

where $\boldsymbol{\delta}$ is the Kronecker *unit tensor* for indices, such that $\delta_{\mathrm{ij}} = 1$ for $i = j$, $\delta_{\mathrm{ij}} = 0$ for $i \neq j$.

Energy Equation: *(from system temperature balance)*

$$\rho C_{\mathrm{p}} \frac{DT}{Dt} = \bigtriangledown \cdot (\lambda \bigtriangledown T) + \beta T \frac{Dp}{Dt} + \phi \tag{4}$$

using the *coefficient of thermal expansion*, β:

$$\beta = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right) \tag{5}$$

and the *dissipation function*, ϕ:

$$\phi = \bigtriangledown \cdot (\tau \overrightarrow{v}) - \overrightarrow{v} \bigtriangledown \cdot \tau \tag{6}$$

where: ρ is the plume density; t is the physical time; $\overrightarrow{v}$ is the plume velocity; $\overrightarrow{f}$ is the body force; p is the plume pressure; $\boldsymbol{\tau}$ is the viscous stress tensor; μ is the first coefficient of viscosity; $\dot{\varepsilon}$ is the *rate of dilatation*; C_p is the specific heat capacity for constant pressure; T is the plume temperature; λ is the

thermal conductivity; β is the coefficient of thermal expansion; and ϕ is the dissipation function or rate of dissipation. A term with a superscript arrow denotes a vector quantity, a term with a superscript dot represents a *flux*, and a term of the form D/Dt indicates a total derivative property. The derivation and use of Equations 1 - 6 are not the focus of this work, hence they will not be developed or explored any further. A complete explanation of their origin and relevance can be found in *Boundary Layer Theory* by Schlichting and Gersten, available in Reference [31]. The use and numerical implementation of the NS equations is described with clarity in *Computational Fluid Dynamics for Engineers* by Cebeci *et al.*, which can be found in Reference [32]. Note that the use of Equation (1) introduces the assumption that system mass is conserved, i.e. mass flux entering or entrained by the system is balanced by an equivalent mass flux leaving the system. Similarly, the use of Equations (2) and (3) implies that the second component of viscosity is negligible compared to the first.

3.2 2D Boundary Layer-Corrected Mach 5 Nozzle

Based on "Design of a Supersonic Nozzle" by McCabe, 1967

3.2.1 Nozzle Geometry

The 2-D x-y co-ordinate data representing the nozzle profile published by McCabe were collected and plotted to check for any errors introduced during manual import and conversion from inches to millimetres. These data, which are reproduced in Appendix A, consisted of only 57 co-ordinate points, which was insufficient to generate smooth Computer-Aided Design (CAD) nozzle geometry for use in a CFD simulation. A *Makima* cubic interpolation scheme was created for coordinate data using *Mathworks MATLAB*, a matrix-based numerical computing language, and used raise the number of co-ordinate points from 57 to 500. The interpolated nozzle profile is illustrated below in Figure 12.

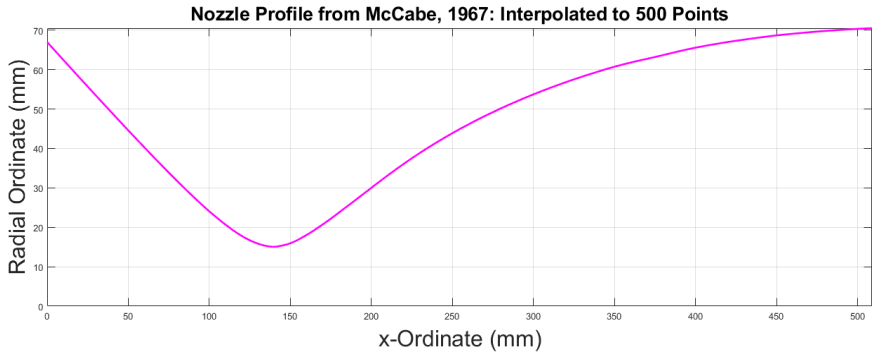


Figure 12: McCabe Boundary-Layer-Corrected Mach 5 Nozzle: Interpolated Nozzle Profile

Essential nozzle properties were evaluated using the same *MATLAB* script, available overleaf in Figure 13, including the radii and areas at the nozzle inlet, throat, and outlet. The nozzle expansion ratio, the ratio

between the nozzle throat and outlet, was determined as approximately 22. Using isentropic relations, the corresponding nozzle exit Mach number was computed as approximately 4.84 for use in informing the development of the associated numerical flow simulation.

```
******** ENG5041: McCabe Nozzle Interpolation and Analysis ********

                 Inlet          Throat          Exit
Area (mm^2):   14083.420       711.289        15617.983
Radius (mm):      66.954        15.047           70.508

Expansion Ratio: 21.96
Nozzle Length (mm): 508.80

>>
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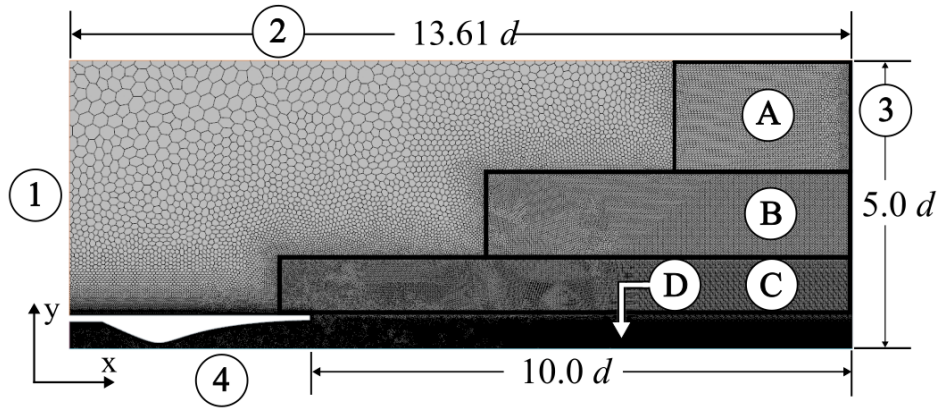
Figure 13: McCabe Boundary-Layer-Corrected Mach 5 Nozzle: Nozzle Properties for Interpolated Geometry

3.2.2 Computational Mesh

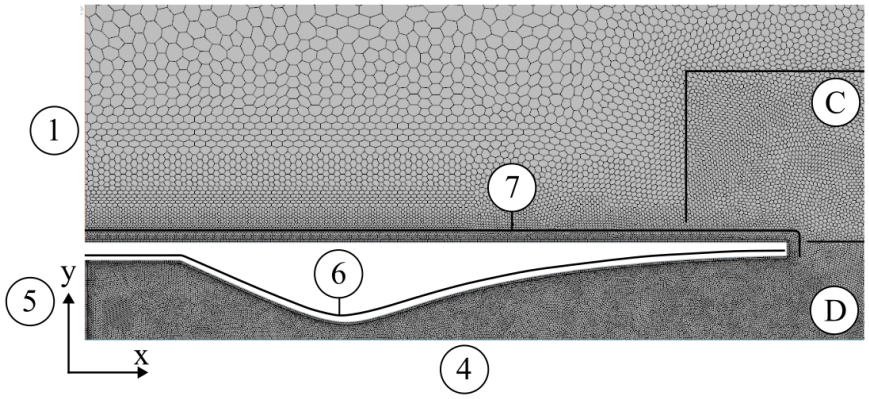
The interpolated x-y co-ordinate data were used to generate solid body geometry, which was the basis of the synthesised computational mesh. An arbitrary dimension of $10d$ was prescribed for the distance between the nozzle exit plane and impingement plate and for the height of the impingement plate. As before, d is the nozzle exit diameter, employed as a geometric normalising parameter. The goal of this simulation was to accurately reproduce flow through the nozzle, thus the plume and impingement flows were not the primary concern; this permitted the use of approximate dimensions for the domain. Planar symmetry was leveraged to maximise computational efficiency by aligning the nozzle axis with the domain x-axis, affording a half-height impingement plate. Overall domain dimensions, including the nozzle body, were 13.61 x 5.0 d (length x height). A small chamber of arbitrary length $0.5d$ was included.

The mesh was developed iteratively, and consisted of a coarse 2-D polygonal surface mesh with superimposed regions of refined mesh to capture expected flow effects. The mesh base size was set at 7.0 mm, with the cell surface growth rate reduced to 1.07 to induce smooth transitions between cell sizes. The remaining default mesh controls were left unchanged. Critical flow features were expected to occur along the plume length [23] [22], at the impingement site [25] [26], and at the interface between the two. Accordingly, rectangular zones of reduced mesh base size were placed to develop a comparatively fine mesh in these areas. Automated mesh refinement toolsets were available, however, manually defining areas of refined mesh yielded greater control. The result was a computationally-efficient mesh that was capable of capturing in detail the flow features of interest, with fine cells where required and comparatively coarse cells elsewhere. The general arrangement of the computational mesh is depicted in Figures 14a and 14b

overleaf, with geometry faces and refinement zones indicated. Tabulated below these figures in Table 1 are the various components and mesh controls corresponding to Faces 1 - 7.



(a) Computational Mesh, Domain



(b) Computational Mesh, Nozzle Treatment

Figure 14: 2-D Boundary-Layer-Corrected Mach 5 Nozzle: Faces 1-7 and Refinement Zones A - D Highlighted

Table 1: 2D McCabe Nozzle Flow Simulation: Base Mesh Controls for Faces 1 - 7

Face(s)	Component	Target Surface Size (% of base size)	Minimum Surface Size (% of base size)	Prism Layer Controls
1	Free Stream	500	Default	Disabled
2	Free Stream	500	Default	Disabled
3	Impingement Plate	50	Default	Default
4	Symmetry Plane	n/a	n/a	n/a
5	Nozzle Inlet	15	7.5	2 Layers; Default Total Thickness
6	Nozzle Internal Surfaces	15	7.5	2 Layers; Default Total Thickness
7	Nozzle External Surfaces	20	10	Default

The data in Table 1 correspond to the base mesh produced. Additional mesh controls superimposed using

regions of refinement A - D are detailed below in Table 2.

Table 2: 2D McCabe Nozzle Flow Simulation: Mesh Refinement Zones A - D

Refinement Zone	Target Surface Size (% of base size)
A	90
B	75
C	50
D	25

A mesh sensitivity analysis was omitted due to the basic nature of this simulation.

3.2.3 Physics Models and Boundary Conditions

Symmetry was leveraged to maximise computational efficiency, which required a 2-D axisymmetric spatial model to be equipped. Uniform flow was enforced numerically to compensate for the short chamber length of only $0.5d$, which would otherwise be insufficient for flow to fully develop. The simulation converged best for an implicit unsteady time-stepping scheme with coupled energy and flow solvers applied. Simulation properties were specified to represent the Martian environment to test this capability for future use, including a reduced-order model of the Martian atmosphere at low altitude developed by Davis *et al.* which was applied across the domain by prescribing a multi-component gas model of 2% argon (Ar), 2% nitrogen (N_2), and 96% carbon dioxide (CO_2) [33]; gas components were assumed non-reactive and ideal. Initial domain physics conditions were informed from scaling parameters derived by Pohly *et al.* (Available in Reference [34]) between Earth sea-level and Mars surface atmospheric conditions. Plume flow was represented as pure nitrogen to force the correct ratio of specific heats [16], denoted γ, between the gas plume and quiescent atmosphere. Impinging flows are, in general, turbulent [4] [16] [2], thus an appropriate turbulence regime was specified. The standard Spalart-Allmaras model of turbulence was optimal for this purpose due to frequent and widespread use in aerospace applications [32], and from prior work, it was known to converge well and possess generous tolerance for flawed mesh geometry. Nozzle inlet design conditions for the McCabe nozzle were not published; incidentally, the MSL MLE nozzle (see Section 2.3.2) delivered a similar nozzle pressure ratio to the McCabe nozzle, thus chamber plenum conditions for the MSL nozzle [16] were scaled against nozzle exit diameter, d, and applied to the McCabe nozzle. The inlet supersonic static pressure was computed using isentropic relations, and domain outlets were given the same physics conditions as the domain medium to ensure free-stream behaviour. Referring back to Figures 14a and 14b for geometry face designations, the boundary and physics conditions applied to each face are given overleaf in Table 3. The simulation was advanced in time until the simulation residuals converged to a quasi-steady state.

Table 3: 2D McCabe Nozzle Flow Simulation: Per-Face Boundary and Physics Conditions

Face(s)	Boundary Condition	Physics Conditions	Properties
1	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO2, N2]	600; 300; [0.2, 0.96, 0.2]
2	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO2, N2]	600; 300; [0.2, 0.96, 0.2]
3	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
4	Axis	Symmetry Plane	n/a
5	Stagnation Inlet	Species Mass Fraction, [Ar, CO2, N2]; Supersonic Static Pressure, Pa; Total Pressure, Pa; Total Temperature, K	[0.0, 0.0, 1.0]; 11420.1; 5E+6; 300
6	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
7	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable

3.3 2D MSL MLE Nozzle

Based on "Thruster Plume Surface Interactions: Applications to Spacecraft Landings on Planetary Bodies" by Mehta et al., 2013

Physical parameters published in the work pertaining to the quarter-scale *SkyCrane* Mars Space Laboratory (MSL) Mars Landing Engine (MLE) simulation were collated for use in numerical method development. For convenience and reference, these data are reproduced in Table 4 below.

Table 4: Quarter-Scale MSL MLE Physical Parameters, Courtesty of Mehta *et al.*, 2013.

Parameter	Symbol	MSL MLE: 100% Throttle, Quarter-Scale
Min. Area, m^2	$A*$	6.9E-5
Exit Area, m^2	A_e	0.00193
Expansion Ratio	$\varepsilon = A_e/A*$	28.0
Chamber Temperature, K	T_c	300.0
Chamber Pressure, kPa	P_c	1765
Exit Pressure, Pa	P_e	2837
Exit Temperature, K	T_e	47
Exit Mach Number	M_e	5.1
Normalised Altitude ^{o}	h/d	34.5
Specific Heat Ratio	γ	1.4

3.3.1 Nozzle Geometry

Geometry for the MSL MLE nozzle was not given in *Thruster Plume Surface Interactions: Applications for Spacecraft Landings on Planetary Bodies*, though nozzle throat and exit areas were prescribed. Using these parameters and the parabolic nozzle curve approximation developed by Rao (available in Reference [35]) for a near-optimum-thrust supersonic nozzle, suitable replacement geometry was obtained. Incorporating the appropriate length and angle approximations [30], an 80% length, 15° half-angle parabolic Convergent-Divergent (CD) nozzle was designed for a nozzle expansion ratio, ε, of 28 and an exit Mach number, M_e, of 5.1. However, initial testing of this nozzle illuminated an incompatibility of the curvature of the divergent section with the flow conditions specified by Mehta *et al.*, producing an erroneous region of accelerated flow immediately downstream of the nozzle exit plane. It was understood that this phenomenon was the consequence of improperly reflected Mach waves converging beyond the nozzle exit, the result of the nozzle operating under off-design conditions [23] [29].

The nozzle design was simplified to eliminate this obstacle, transforming the geometry from parabolic to conical whilst retaining the nozzle length and angles. This simplification was valid to make, as conical nozzles are frequently used in conjunction with spacecraft propulsive devices [30], and the nozzle geometries considered were sufficiently small in dimension to render the inevitable reduction in nozzle efficiency [35] negligible. The ultimate nozzle profile is given below in Figure 15, depicted as the inverse geometry for use in a numerical simulation. The short chamber length of only 0.5 d, which is generally insufficient for flow to fully develop, would later be made possible in this application by means of a mathematical field function programmed to ensure developed flow at the plenum exit.

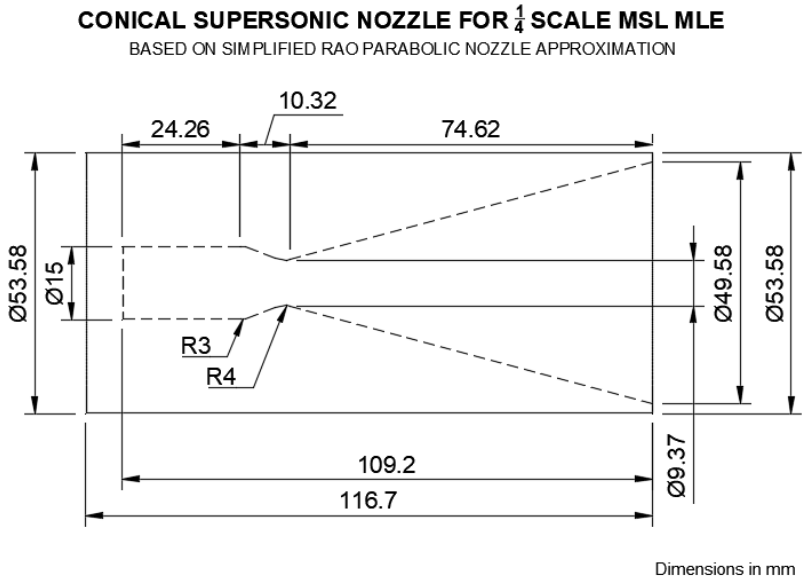


Figure 15: 2D MSL MLE Simulation: Conical Nozzle Geometry

3.3.2 Simulation Domain

Mehta *et al.* specified a domain size of several hundred d in the x, y, and z directions, which was prohibitively computationally expensive for this study. The domain volume was reduced more than ten-fold by replacing the 3-D spatial model implemented by Mehta *et al.* with a 2-D axisymmetric spatial model, where the nozzle axis was aligned co-axially with the domain x-axis. This simultaneously removed the requirement to simulate flow in the z-direction (domain depth) and exploited the 2-D planar symmetry of the nozzle to halve the simulated 2-D plane about the nozzle or plume axis. Note that the exchange of a 3-D spatial model for a 2-D model was valid for nozzle flow only in the case of an axisymmetric model, which effectuated a revolve of the nozzle to preserve its throat and exit areas, and hence its expansion ratio, ε. Had a 2-D planar model been equipped, the nozzle expansion ratio would have been a function only of the throat and exit radii, reducing the expansion ratio, ε, from 28.0 to approximately 4.0, and producing a nozzle exit pressure, P_e, of the order of several gigapascals. This highly-efficient spatial modelling strategy incurred the cost of no longer being able to model the 22.5° angle of inclination of the landing engines due to the required alignment of the nozzle axis with the domain x-axis. However, such a change in the simulation arrangement was more than acceptable, as the focus of the simulation was to model the general far-field flow physics exhibited by a generic impinging plume.

Evident from the extent of the flow in the published Mach contours, included in Figure 10a in Section 2.3.2, an impingement plate width of around 40 d was sufficient to capture radially-propagating wall flow. Of this width, only half had to be represented due to the equipped spatial model. The domain space above the simulated nozzle geometry was convenient to reduce, as an expanding jet in a very thin atmosphere was not expected to entrain a significant amount of free-stream flow; the length of the nozzle itself was thought a sufficient buffer between the nozzle exit and free-stream flow boundary. A further reduction was made via the tapering of the domain space at the outermost corner from the impingement point, as flow here was expected to be entirely quiescent and thus would be unproductive to simulate. The consequent 2-D domain was 36.7 d in width and 20.0 d in height, and featured a taper in the top-left (north-west) corner of the domain space as illustrated overleaf in Figure 16. The inverse geometry of the conical nozzle was fused with the domain grid. To comply with axisymmetric model requirements, the domain was rotated in sense about the z-axis (depth axis) by 90° anticlockwise compared to the Mach contours provided by Mehta *et al.*, reproduced in Figure 10a in Section 2.3.2. Although much smaller than the simulation domain developed by Mehta *et al.*, the 2-D domain was highly computationally efficient due to its small size whilst still capturing the flow effects of interest. Greater domain distances would be required to allow flow to develop if an earth-similar atmosphere was particularised, as flow effects propagate to a greater extent for substantive atmosphere [3].

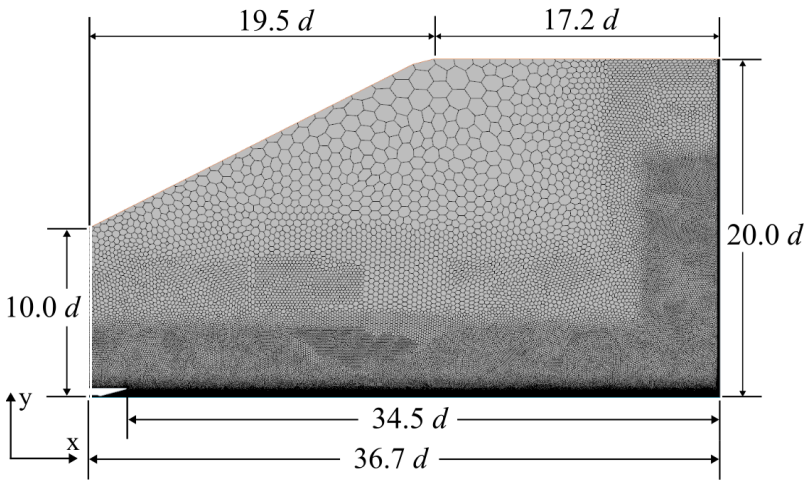
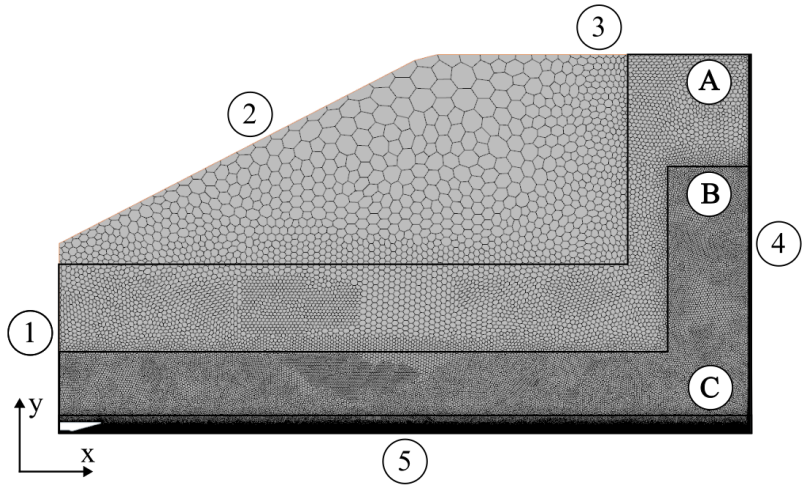


Figure 16: 2D MSL MLE Simulation: Simulation Domain Arrangement

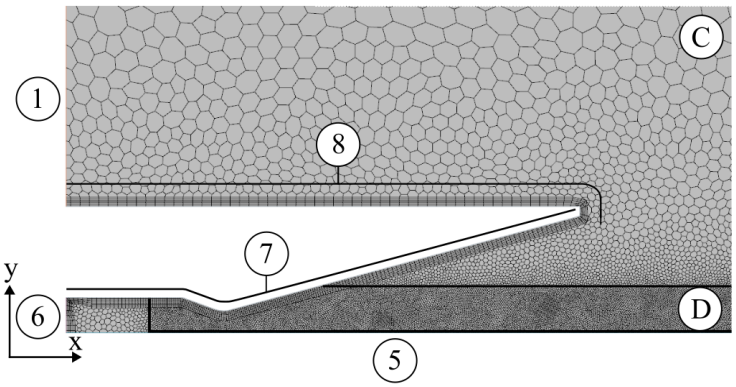
3.3.3 Computational Mesh

A computational mesh was generated using an iterative process, which began with a 2-D polygonal surface mesh of base cell size 0.02 m and a surface growth rate of 1.1 to synthesise a simple and coarse mesh across the whole domain. The remaining default mesh controls were unmodified. Surface controls were imposed on domain geometry faces to develop the mesh local to these areas, detailed in Table 5 overleaf for each face. Below in Figure 17a and overleaf in Figures 17b and 17c are schematics of the domain, indicating the designations of faces for use with Table 5. A thick prism layer was specified at the impingement plate to capture viscous effects influencing the radially-propagating wall flow. Prism layers were also applied to the nozzle's internal and external surfaces to capture the boundary layers in these areas.

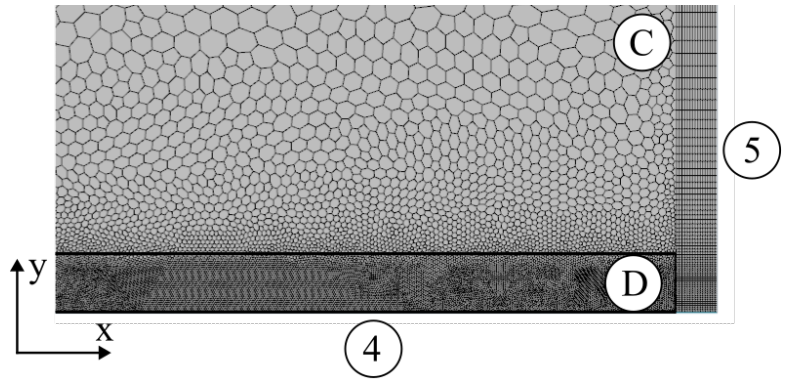


(a) Domain View

Figure 17: 2D MSL MLE Simulation, Computational Mesh: Geometry Faces and Prescribed Areas of Mesh Refinement Indicated



(b) Nozzle Treatment



(c) Impingement Plate Treatment

Figure 17: 2D MSL MLE Simulation, Computational Mesh: Geometry Faces and Prescribed Areas of Mesh Refinement Indicated

Table 5: 2D MSL MLE: Base Mesh Controls for Faces 1 - 8

Face(s)	Component	Target Surface Size (% of base size)	Min. Surface Size (% of base size)	Prism Layer Controls	Prism Layer Properties
1	Free Stream	700	Default	Disabled	n/a
2	Free Stream	700	Default	Disabled	n/a
3	Free Stream	700	Default	Disabled	n/a
4	Impingement Plate	50	Default	Layers; Total Thickness (% of base size);	16; 40
5	Symmetry Plane	n/a	n/a	Disabled	n/a
6	Nozzle Inlet	5	1	Layers; Total Thickness (% of base size);	7; 10
7	Nozzle Internal Surfaces	5	1	Layers; Total Thickness (% of base size);	7; 10
8	Nozzle External Surfaces	10	5	Layers; Total Thickness (% of base size);	7; 10

The polygonal primary meshing scheme made large changes in cell size possible without compromising the integrity of the mesh, as the cell shape allowed for fluid changes in mesh size over a short distance. Polygonal cells also transmit data to a greater number of neighbouring cells than alternative meshing schemes whilst avoiding negative cell qualities such as high aspect ratios or skewed cells, and are well-suited to high-speed compressible flows [33] [32].

Complementary to the method employed by Mehta *et al.*, a composite of a block-structured mesh and an unstructured mesh was achieved by prescribing rectangular areas of reduced cell size at locations where flow phenomena were expected to improve the local mesh quality. As before, these locations were along the plume length [23] [22] and at the impingement site [25] [26], where manually applying regions of improved mesh quality gave superior control over automated mesh refinement tools. The cell sizes specific to each region of refinement were selected such that mesh quality reduced with increasing distance from the impingement plate and plume centre-line, intended to stage the gradual transition between small and large cells. The four refinement zones, denoted *A*-*D*, and their associated mesh controls are detailed in Table 6 below. The locations and extent of refinement zones *A*-*D* are illustrated in Figures 17 prior.

Table 6: Mesh Refinement Zones A - D for 2D MSL MLE Simulation

Refinement Zone	Target Surface Size (% of base size)
A	60
B	30
C	15
D	2

The completed mesh was computationally economical despite containing multiple regions of high-quality mesh, as these regions were placed strategically to capture expected flow phenomena whilst keeping other parts of the mesh comparatively coarse. Similar mesh grids were generated for base sizes of 0.01 and 0.03 m to facilitate a sensitivity analysis correlating varying mesh quality and the respective results obtained.

3.3.4 Physics Models and Boundary Conditions

Matching the physics models selected by Mehta *et al.*, the simulation was solved across the entire domain using the full Reynolds-Averaged Navier-Stokes (RANS) equations, where gas flow was assumed to behave according to the ideal gas model. System mass conservation was assumed, and the simulation converged in the shortest physical time and was most stable for a steady-flow time-marching scheme. Applying the flow physics models detailed in the study, a k-Ω turbulence regime was equipped, including gas compressibility correction and Menter's Shear-Stress-Transport (SST) model. From experience, this turbulent modelling

approach was known to produce accurate results for high-speed flows. A grid sequencing initialisation scheme was applied to eliminate temperature singularities encountered in the nozzle chamber during the initial few-hundred simulation iterations, and a continuity convergence accelerator configured for high-speed compressible flows was incorporated to reduce the simulation Total Computational Time (TCT). Wall boundaries were modelled with an infinite y+ factor to force viscous spacing. A total-pressure, total-temperature stagnation inlet boundary condition was used to satisfy nozzle inlet conditions. Nozzle chamber conditions detailed in Table 4 were prescribed at the plenum, with the nozzle supersonic static pressure computed using essential isentropic relations.

The Martian atmosphere was represented using a reduced-order atmospheric model developed by Davis *et al.*, given in Reference [33], comprised of 2% argon (Ar), 2% nitrogen (N_2), and 96% carbon dioxide (CO_2). Gas components were modelled as coupled and non-reacting. Mehta *et al.* specified initial conditions of 690 Pa and 290 K, which were applied to both the domain and the outlet boundaries to maintain free-stream conditions in these regions. Omitted atmospheric data were informed using scaling parameters between Earth and Mars atmospheres, detailed in Reference [34]. The plume gas was modelled as only nitrogen to force the correct specific heat ratio of $\gamma = 1.4$ between the ambient and plume flows. Strict minimum temperature and pressure controls of 50 K and 50 Pa respectively were applied at the initialisation of the simulation to aid convergence, and were gradually relaxed to 1 K and 1 Pa as the simulation progressed. Boundary and physics conditions are tabulated below in Table 7, where the face designations can be recalled by reviewing Figures 17.

Table 7: 2D MSL MLE Simulation: Per-Face Boundary and Physics Conditions

Face(s)	Boundary Condition	Physics Conditions	Properties
1	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO2, N2]	690; 290; [0.2, 0.96, 0.2]
2	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO2, N2]	690; 290; [0.2, 0.96, 0.2]
3	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO2, N2];	690; 290; [0.2, 0.96, 0.2]
4	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
5	Axis	Symmetry Plane	n/a
6	Stagnation Inlet	Species Mass Fraction, [Ar, CO2, N2]; Supersonic Static Pressure, Pa; Total Pressure, Pa; Total Temperature, K	[0.0, 0.0, 1.0]; 2828.27; 1675E+3; 300
7	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
8	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable

The simulation was allowed to run until the simulation residuals had reached a quasi-steady condition and had converged to acceptably small magnitudes.

3.4 3D Supersonic Fluidic Injector Simulation: Flow Off

Based on "Effectiveness of a Supersonic Jet-Jet Interaction Technique on Flow Structure in an Underexpanded Jet Impinging on a Perpendicular Plate" by Kobayashi and Ukai, 2022

3.4.1 Simulation Arrangement and Nozzle Geometry

The relative positions and dimensions of experimental components described by Kobayashi and Ukai, available in Reference [9], were collected and scaled against the exit diameter of the primary nozzle, d. Chamber conditions for the supersonic fluidic injectors were scaled against the primary nozzle chamber pressure and temperature, P_c and T_c respectively, and the exit Mach number of the injectors was defined as a ratio of the primary nozzle exit Mach number. Primary nozzle geometry was required; these scaled data were applied to the quarter-scale MSL MLE nozzle geometry (Available in Figure 15) as by this point its use and behaviour were well understood, which would aid in future validation of this simulation. Kobayashi and Ukai gave inlet, throat, and exit areas for the supersonic fluidic injectors used, and applying the simplified (conical) Rao curve approximations developed for the MSL MLE simulation, including the same 80%-length and 15° half-angle assumptions, the following injector nozzle geometry was obtained:

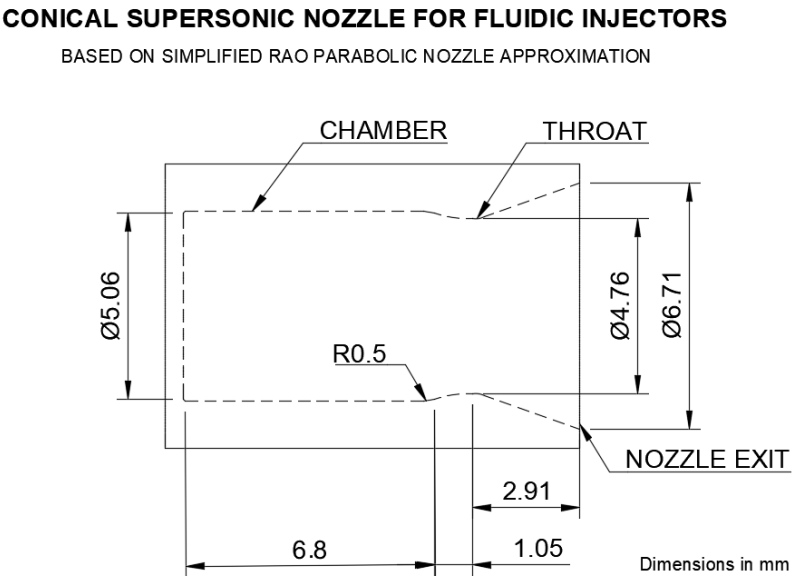


Figure 18: Supersonic Injector Simulation: Injector Nozzle Geometry

Using isentropic relations and the computed area ratio for this geometry of 1.988, the injector exit Mach number was calculated as 3.855. The general arrangement of this simulation is given overleaf in Figure

19, where the conical MSL MLE nozzle geometry was unaltered from previous work:

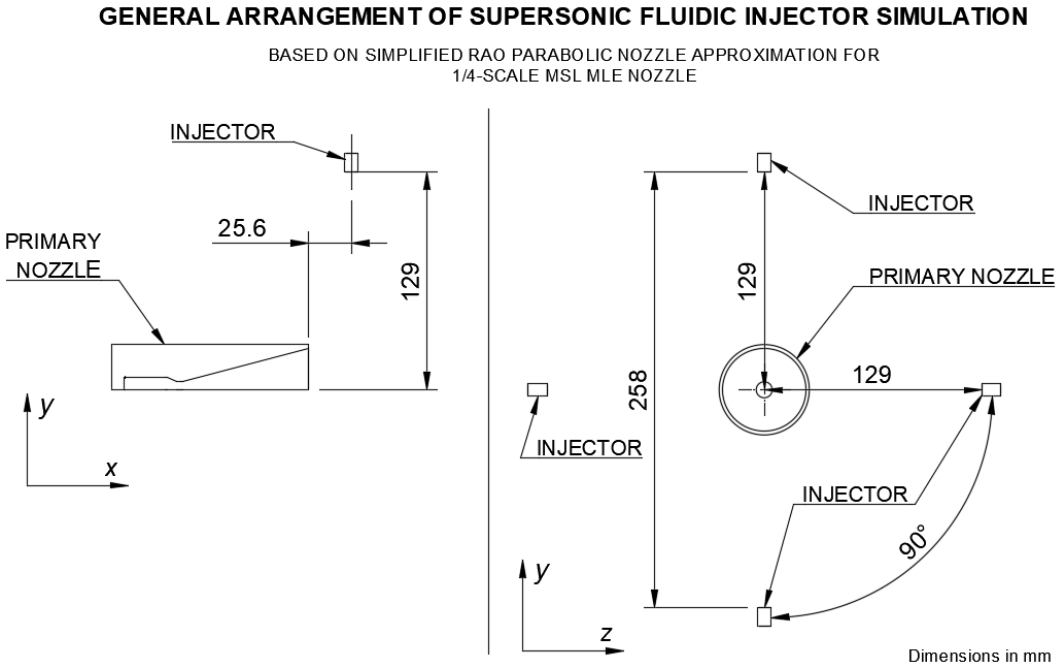


Figure 19: Supersonic Injector Simulation: General Arrangement

Note that the left panel in the above figure depicts a quarter-revolve of the nozzle in the x-y plane, and the right shows a full-revolve of the nozzle looking along the nozzle axis in the z-y plane. The inverse geometries of the primary and injector nozzles are given above, configured for use in a CFD simulation.

3.4.2 Simulation Domain

Leveraging symmetry in the goal of maximising computational efficiency, only a quarter-revolve of the inverse nozzle geometry was simulated, with only one of the four injectors included. To reproduce the arrangement given in Figure 19 using only a 90° revolution, the injector was placed at 45° clockwise from the x-y plane. Kobayashi and Ukai specified a maximum impingement distance of $10d$ with an impingement plate dimension of 13.87 x 20.91 d, however, the plate was increased in overall size to 20 x 20 d as the MSL MLE nozzle possessed a greater exit Mach number than the nozzle employed by Kobayashi and Ukai, and thus its radially-propagating flow may have spanned a larger area. Similarly, only the maximum impingement altitude of $10d$ was reproduced. Simulating 3-D flows is highly computationally intensive, so the domain was generated to avoid expending needless computational effort in simulating regions of quiescent flow, and to reduce the computational load where possible. To that effect, the domain was synthesised using a revolved sketch of conical form, which tapered towards the nozzle where a cylindrical section of the domain was specified to capture whatever minor flow entrainment effects

occurred at the upstream domain interface, and also to house the injector and its plume flow. Including the length of the nozzle body, the resulting domain was $11.97d$ in length, with an impingement plate of radius $10d$ simulated across a 90° revolution, detailed below in Figure 20 for the 2-D revolved plane.

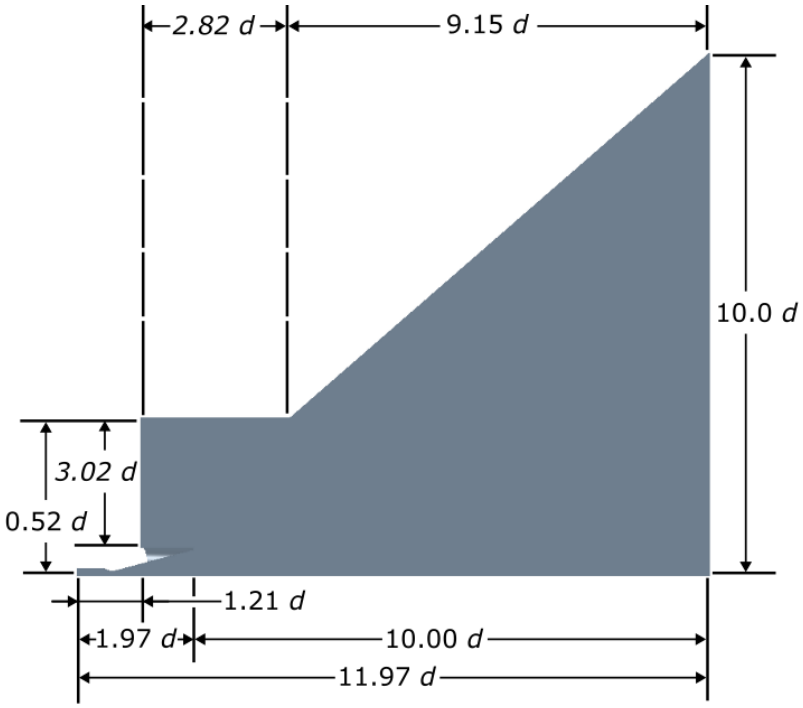


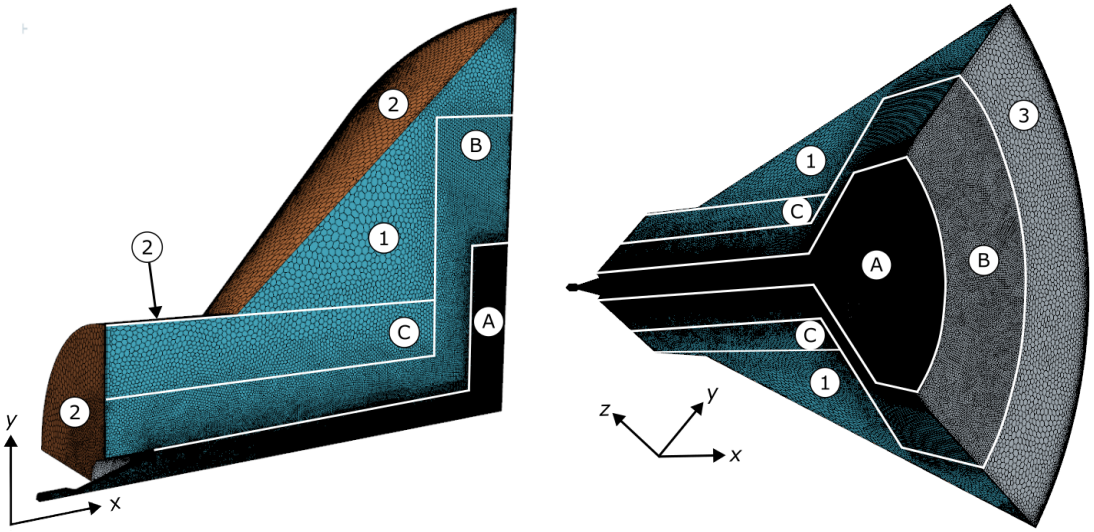
Figure 20: Supersonic Injector Simulation: Domain Arrangement

3.4.3 Computational Mesh

The computational mesh was generated in an iterative process, starting with a basic, coarse mesh synthesised using a prescribed base size of 9.0 mm and surface growth rate of 1.05, where the remaining default controls were left unaltered. A polyhedral mesher was equipped for surface meshing to produce fluid changes between cell sizes, coupled with an advancing layer volumetric meshing scheme due to proven ability to produce good quality results for high speed flows [32] and capture boundary layer flow.

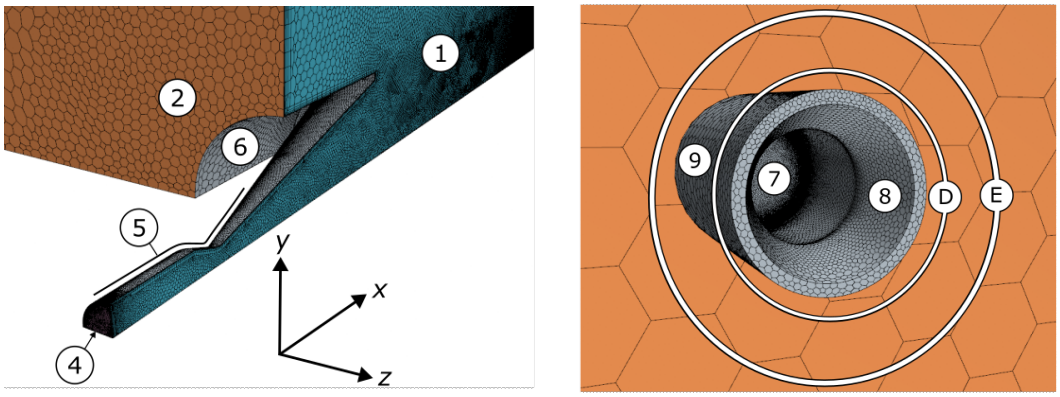
To improve the surface mesh quality local to surfaces of interest, such as the nozzle inlet or impingement plate, a series of surface mesh controls were applied to domain faces. Manually prescribed volumetric regions of refined mesh were superimposed to strike a balance between an unstructured and block-structured mesh, where these regions possessed a smaller cell size than the surrounding domain, following the approach similar to that developed by Mehta *et al.* used in the prior MSL MLE simulation. This approach yielded a computationally efficient mesh with a refined mesh grid incident where interesting flow effects were expected, and a comparatively coarse mesh elsewhere, where the superimposed volumetric refinement produced smooth changes in cell size across the domain. Prism layers were applied to the mesh

grid to capture boundary layer flow in the nozzles and across the impingement plate. Illustrated below in Figures 21a to 21d is the developed mesh grid, with Surfaces 1 - 9 and Refinement Zones *A* - *E* highlighted and the domain orientation indicated.



(a) Domain View

(b) Impingement Plate



(c) Primary Nozzle

(d) Injector Nozzle

Figure 21: Supersonic Fluidic Injector Simulation: Computational Mesh

Note that the supersonic fluidic injector nozzle depicted in Figure 21d is *inside* the mesh grid, where the orange background is in fact the internal surface of the outlet wall, denoted Surfaces 2. The concentric rings marked *D* and *E* are the radii of cylindrical regions of volumetric refinement applied along the injector nozzle axis, which extended from the injector nozzle exit plane to the domain axis. Refinement Zones *A* - *C* existed inside the revolved geometry as well as on the surface, in the sense indicated. The prescribed surface mesh controls and the volumetric mesh controls applied by the specified regions of refinement are available below in Tables 8 and 9 respectively, where the surface and zone designations correspond to those given in Figures 21 previous.

Table 8: Per-Face Mesh Controls for 3D Supersonic Fluidic Injector Simulation

Surface(s)	Component	Target Surface Size (% of base size)	Min. Surface Size (% of base size)	Prism Layer Controls	Prism Layer Properties
1	Symmetry Plane	120	10	Disabled	n/a
2	Free Stream	100	Default	Disabled	n/a
3	Impingement Plate	90	Default	Layers; Layer Stretching; Total Thickness (% of base size);	7; 1.5; 33.33
4	Primary Nozzle Inlet	2.5	1.25	Layers; Total Thickness (% of base size);	5; 10
5	Primary Nozzle Internal Surfaces	7.5	1.25	Layers; Total Thickness (% of base size);	5; 15
6	Primary Nozzle External Surfaces	30	15	Layers; Total Thickness (% of base size);	5; 20
7	Injector Nozzle Inlet	1	0.5	Layers; Layer Stretching; Total Thickness (% of base size);	3; 2.0; 7
8	Injector Nozzle Internal Surfaces	7.5	0.25	Layers; Layer Stretching; Total Thickness (% of base size);	3; 2.0; 10
9	Injector Nozzle External Surfaces	30	15	Layers; Total Thickness (% of base size);	3; 15

Table 9: Mesh Refinement Zones A - E for 3D Supersonic Fluidic Injector Simulation

Refinement Zone	Target Surface Size (% of base size)
A	10
B	35
C	70
D	10
E	40

3.4.4 Physics Models and Boundary Conditions

To test this new simulation and to provide a good benchmark for the effectiveness of a supersonic jet-jet interaction, no flow was prescribed through the injector nozzle. This also served as an adequate assessment of the expansion of the simulation method into 3-D, as the MSL MLE nozzle had previously only been represented in 2-D. In keeping with previous work and again informed by the numerical simulation developed by Mehta *et al.*, the simulation was solved using the full RANS equations across the whole domain, using an ideal gas model with conservation of system mass assumed. Superior stability was observed for a steady time-stepping scheme, and as before, a k-Ω turbulence model was equipped, with Menter's Shear-Stress-Transport (SST) model and automated compressibility corrections. This combina-

tion of physics models was known to give accurate results for compressible, high-speed flows [32], and was applied successfully to the prior MSL MLE simulation. Default coupled flow solvers were applied, and temperature singularities that occurred within the nozzles during the first few-hundred iterations were eliminated using a grid sequencing initialisation scheme. Total Computational Time (TCT) was reduced by equipping a continuity convergence accelerator configured for compressible and high-speed gas flows.

Infinite wall y+ viscous spacing treatment was applied to solid interfaces as specified by Mehta *et al.*, and total pressure, total temperature stagnation inlet boundary conditions satisfied the primary nozzle plenum conditions, where these data and the primary nozzle supersonic static pressure were identical to the previous simulation (see Section 3.3). Similarly, the primary nozzle flow was again represented as nitrogen to yield the correct ratio of specific heats between the plume flow and the surrounding media. Also identical to the simulation prior were the domain initial conditions and the reduced-order model of the Martian atmosphere reproduced in this simulation [33], where gas components were assumed inert. The surfaces at the domain extent were specified as pressure outlets operating with the same properties applied as domain initial conditions, namely a pressure and temperature of 690 Pa and 290 K respectively, representing a free-stream condition for static ambient flow. Note that for this simulation, the supersonic fluidic injector nozzle inlet was modelled as a wall, with no flow prescribed. Convergence was promoted by enforcing strict minimum temperature and pressure restrictions of 60 K and 60 Pa respectively, a value determined iteratively to avoid the introduction of singularities without significantly influencing the simulation results. Applied boundary and physics conditions are summarised below in Table 10, where the surface designations 1-9 are those given in Figures 21 previously.

Table 10: Per-Face Physics Conditions for 3D Supersonic Fluidic Injector Simulation, Flow Off

Surface(s)	Boundary Condition	Physics Conditions	Properties
1	Symmetry Plane	n/a	n/a
2	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO2, N2]	690; 290; [0.2, 0.96, 0.2]
3	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
4	Stagnation Inlet	Species Mass Fraction, [Ar, CO2, N2]; Supersonic Static Pressure, Pa; Total Pressure, Pa; Total Temperature, K	[0.0, 0.0, 1.0]; 2828.27; 1675E+3; 300
5	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
6	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
7	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
8	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
9	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable

An analysis of mesh sensitivity was prepared by running this new simulation for two further mesh grid base sizes of 7.0 and 11.0 mm. All three simulation was advanced in time until the plotted values of the simulation residuals achieved a quasi-steady state. The optimal mesh grid base size determined for the 'injector-off' simulation would later be applied to the subsequent two related simulations, as the simulation without injector flow was the least computationally intensive of the three and it was desirable to maintain the same mesh grid base size across all supersonic fluidic injector simulations. Whilst a sensitivity analysis for each simulation was desirable, the prohibitive computational expense of evaluating such a physically large and complex flow restricted the total number of simulations that could be executed.

3.5 3D Supersonic Fluidic Injector Simulation: Flow On

The nozzle geometries, simulation domain, and computational mesh for this simulation were identical to the last (see Section 3.4), with the only discrepancy being that flow was now prescribed through the injector nozzle according to the scaled parameters described previously. An updated list of simulation physics conditions, the only family of data that were altered, is given in Table 11 below.

Table 11: Per-Face Physics Conditions for 3D Supersonic Fluidic Injector Simulation, Flow On

Surface(s)	Boundary Condition	Physics Conditions	Properties
1	Symmetry Plane	n/a	n/a
2	Pressure Outlet	Pressure, Pa; Temperature, K; Species Mass Fraction, [Ar, CO2, N2]	690; 290; [0.2, 0.96, 0.2]
3	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
4	Stagnation Inlet	Species Mass Fraction, [Ar, CO2, N2]; Supersonic Static Pressure, Pa; Total Pressure, Pa; Total Temperature, K	0.0, 0.0, 1.0]; 2828.27; 1675E+3; 300
5	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
6	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
7	Stagnation Inlet	Species Mass Fraction, [Ar, CO2, N2]; Supersonic Static Pressure, Pa; Total Pressure, Pa; Total Temperature, K	0.0, 0.0, 1.0]; 14.13E+3; 1675E+3; 300
8	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable
9	Wall	Shear Stress; Thermal Condition; Wall Species;	No-Slip; Adiabatic; Impermeable

The expansion ratio of the injector nozzle was determined from its specified geometry (Available in Reference [9] and reproduced in Figure 18) as 1.988. Matching the experimental arrangement outlined by Kobayashi and Ukai, the injector nozzle plenum pressure matched the primary nozzle plenum pressure, simulating siphoned flow from the primary nozzle feeding the injector nozzle(s), at 1765 kPa. From

these data and using isentropic relations, a supersonic static pressure of 14.130 kPa was computed. In keeping with the numerical methodology employed thus far, the injector nozzle flow was prescribed as only nitrogen to force the correct ratio of specific heats between its plume flow and the quiescent domain gas. The primary and injector nozzles fired simultaneously, and once again, the simulation was allowed to progress until simulation residuals achieved a quasi-steady condition.

3.6 3D Canted Injector Simulation

The previous geometry and arrangement of the supersonic fluidic injector simulation (see Sections3.4 and 3.5) was modified to include a 5° clockwise rotation of the injector nozzle about its exit plane, in the sense indicated in Figure 22 below. This angle was carefully selected to provide the maximum possible primary nozzle plume gas swirl velocity whilst ensuring the injector plume still intersected with the primary nozzle plume core, and thus continued to produce a successful jet-jet interaction to eliminate the majority of the incident shock wave structures.

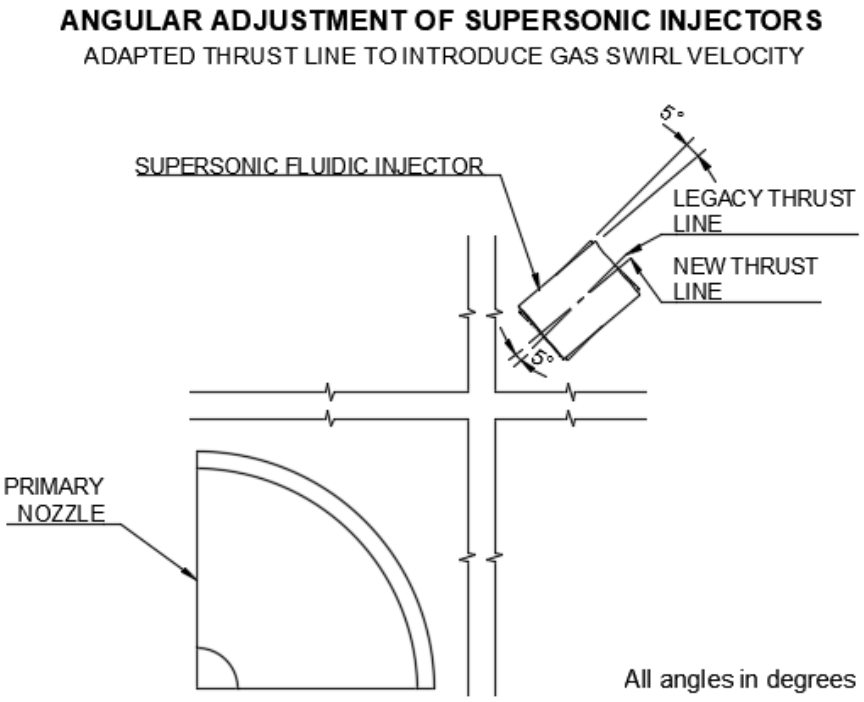


Figure 22: Supersonic Fluidic Injector Simulation: Adjusted Line of Action of Injector Nozzle

The adapted geometry was re-meshed, and the simulation was re-run with identical physics and boundary conditions to the immediately prior simulation. These data are not reproduced here to avoid repetition and are available in Table 11 previous. A final time, this simulation was advanced in time until its residuals had converged to a quasi-steady condition with acceptably small magnitudes.

4 Experimental Results

4.1 2D Boundary Layer-Corrected Mach 5 Nozzle

4.1.1 Mach Contours

Presented in Figure 23 below are the Mach contours obtained from the 2-D simulation of flow through the boundary layer-corrected supersonic nozzle designed by McCabe, intended to function as a test of the validity of the developed 2-D nozzle and near-field flow simulation method. Note that this image depicts the simulation domain mirrored about the nozzle axis or plume centreline for clarity.

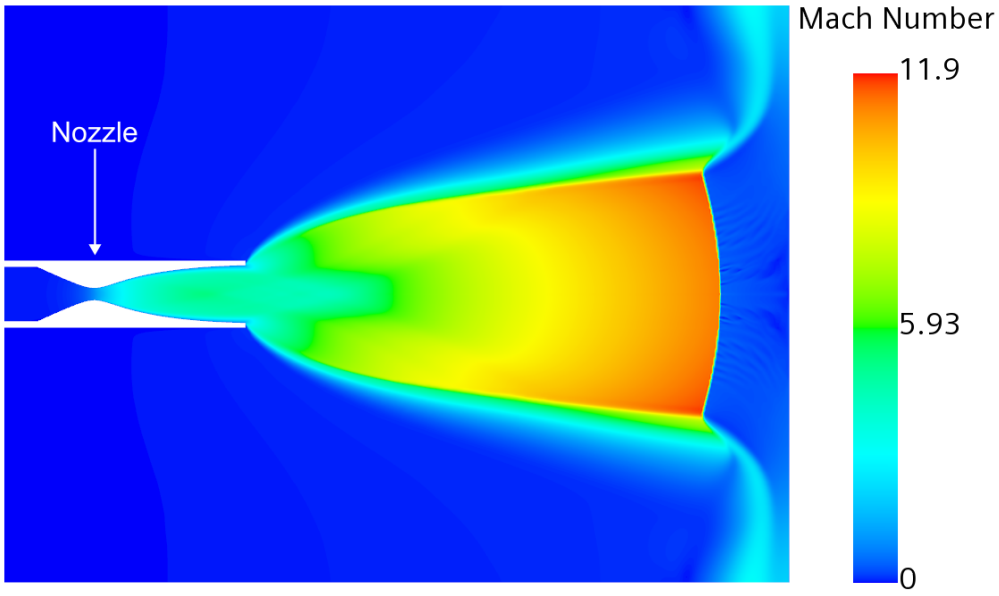


Figure 23: McCabe 2D Boundary Layer-Corrected Mach 5 Nozzle Flow Simulation, Mach Contours

4.1.2 Analysis and Validation

Understood from reviewed literature, physical and correct nozzle flow should exhibit a uniform rate of change of velocity from the nozzle inlet-chamber interface, or the start of the nozzle convergent section, to the nozzle exit plane, or the end of the divergent section [30]. The gas velocity across the nozzle exit plane should also be uniform, and for this nozzle geometry, an exit Mach number of approximately 5 was specified [29]. From the above contours in Figure 23, it can be seen that the flow through the nozzle exhibits uniform gas velocity across the nozzle exit plane, and approximately uniform gas acceleration across the nozzle length from the throat to the exit plane. This is further evidenced overleaf in Figure 24, in which sampled gas velocity within the nozzle is plotted against distance from the nozzle throat, and in Figure 25, which shows a plot of gas Mach number across the nozzle exit plane versus radial distance from the nozzle axis. Considering figure 24, the rate of change, or gradient, of the plotted gas

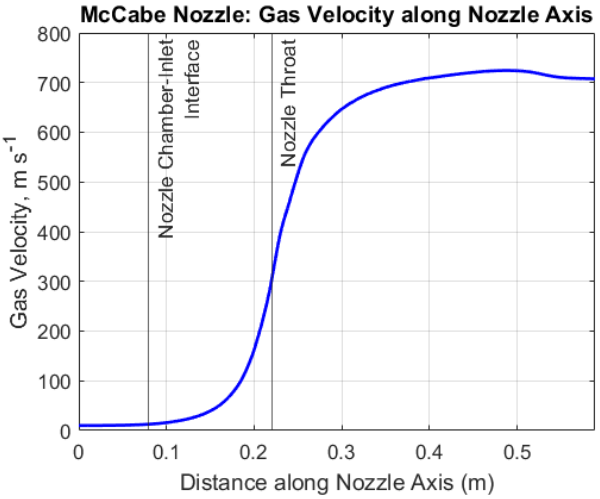


Figure 24: 2D McCabe Nozzle Simulation: Gas Velocity along Nozzle Axis

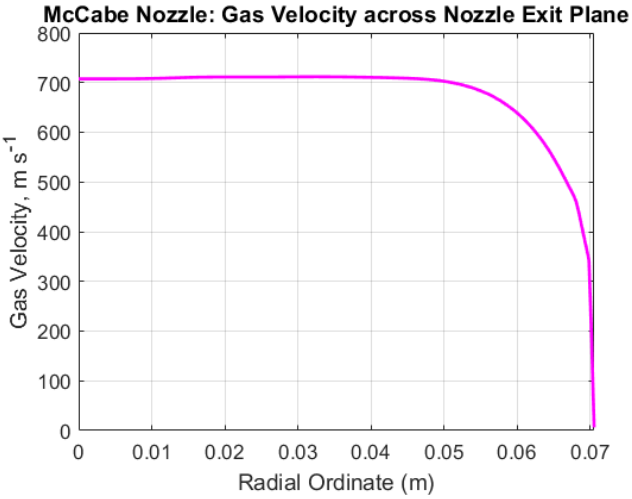


Figure 25: 2D McCabe Nozzle Simulation: Gas Velocity across Nozzle Exit Plane

velocity is predominantly linear, particularly local to the nozzle throat, indicating the expected uniform rate of change of velocity. The plateau towards the nozzle exit plane, located at approximately 0.6 m, is likely the result of the nozzle operating under marginally off-design conditions due to the specification of a gas model representative of a Martian atmosphere, instead of the intended Earth atmosphere. The simulation was modelled for a flow of finite viscosity with viscous effects applied, where the curvature exhibited in Figure 25 is the result of viscous forces having acted to reduce the gas velocity to zero at the nozzle internal wall, located at approximately 0.07 m from the nozzle axis. Additionally, the mean gas velocity of 700 m s^{-1} detailed in the Figure corresponds to a gas Mach number of approximately 5.1, which is acceptably close to the expected value of 5.0. Although not the focus of this simulation, the flow structures at the point of impingement are superficially similar to those detailed in the reviewed literature [16] [12] [25] [26], including a strong bow shock encompassing a region of stagnated flow, a clear shear layer, and a triple point leading to well-defined tail shocks and radially propagating wall jets. These flow features are annotated in Figure 26 overleaf, which depicts a pseudo-schlieren image obtained using a series of algorithms to identify strong changes in gas density, including reflected and intersecting Mach waves. In this figure, the brightest zones indicate the strongest changes in density.

Due to the literature-accurate flow features captured in this simulation and an exit Mach number approximate to the design value, the results produced by the 2-D boundary layer-corrected McCabe nozzle flow simulation were accepted as valid and accurate, and, by extension, so was the simulation method used to produce them. The simulation converged after around 15,000 iterations, where the residuals converged to values between the orders of $x10^{-1}$ and $x10^{-3}$, which was perfectly acceptable for a simulation involving compressible high-speed, unsteady, and non-linear flows.

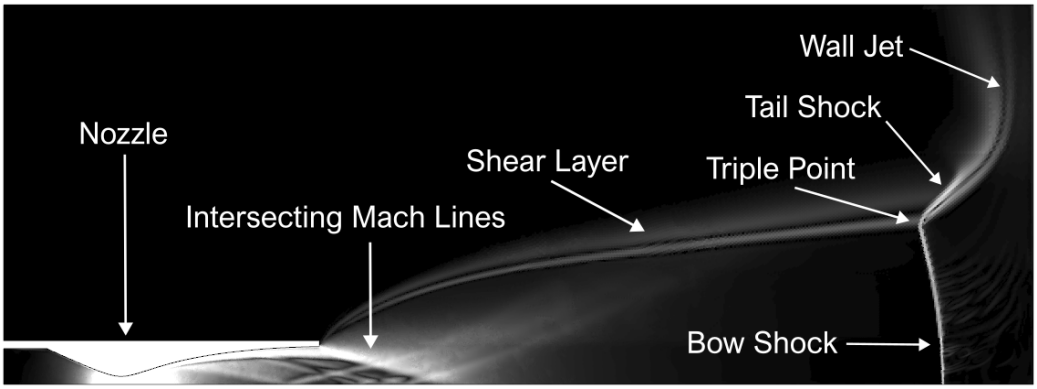
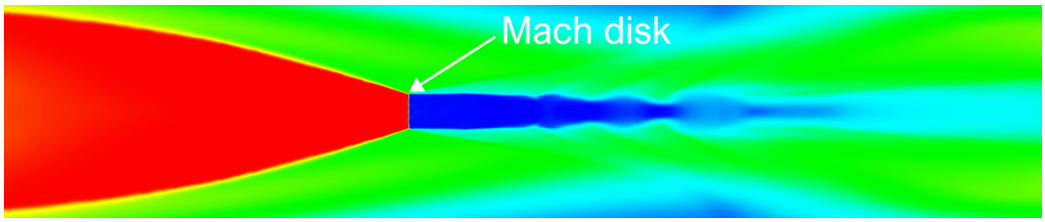


Figure 26: 2-D Simulation of McCabe Nozzle Flow, Pseudo-Schlieren Image

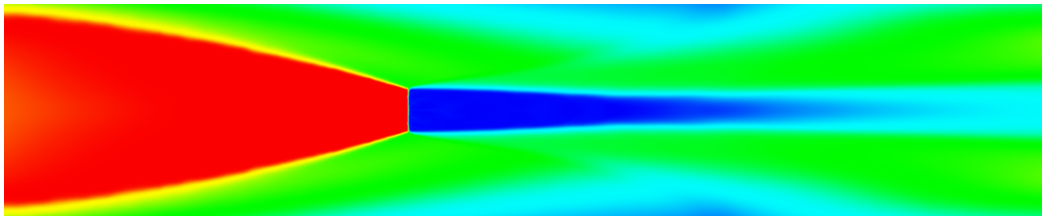
4.2 2D MSL MLE Nozzle

4.2.1 Mesh Base Size Sensitivity Analysis

The simulation was computed for mesh grid base sizes of 0.01, 0.02, and 0.03 m to monitor the influence of base size on simulation results. The effect on the simulated flow of varying mesh base size is illustrated below in Figures 27a and 27b, and overleaf in Figure 27c, which detail plume flow local to a *Mach disk* located at the terminus of the first *shock cell* for each simulated base size. The Mach disk is highlighted in Figure 27a, and a colour bar is supplied overleaf in Figure 27d.



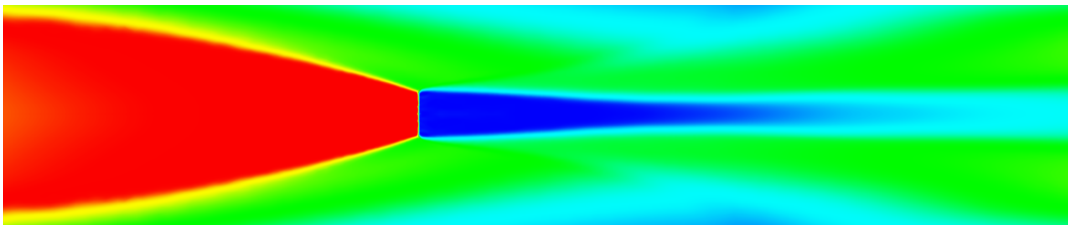
(a) Base Size 0.01 m



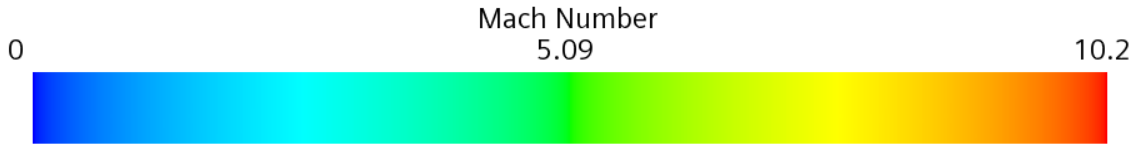
(b) Base Size 0.02 m

Figure 27: 2D Sub-Scale MSL MLE Nozzle Simulation: Sensitivity Analysis, Mach Disc

Conspicuous in Figure 27a is the unstable, turbulent plume flow in the *slipstream* immediately downstream of the Mach disc for the base size of 0.01 m, which did not recover before being diminished by turbulent mixing and viscous effects. Also evident in this figure are the well-defined boundaries between the Mach



(c) Base Size 0.03 m



(d) Colour Bar

Figure 27: 2D Sub-Scale MSL MLE Nozzle Simulation: Sensitivity Analysis, Mach Disc

10 flow within the shock cell, the Mach 7 shear flow region, and the Mach 5 plume flow, where this base size captured complex supersonic flow structures in high resolution. The Mach disc itself was well-resolved, indicated by the imperceptible margin between the shock cell flow and the greatly decelerated flow downstream of the disc. Comparing Figures 27b and 27c, the base size of 0.02 m better resolved the slipstream, disclosed by the straighter and slightly sharper edges detailed in Figure 27b, and the density change across the Mach disc is better defined, indicated by a barely-perceptible margin between the shock cell flow and decelerated flow immediately downstream. Plotted overleaf in Figure 28 is the plume gas Mach number along the plume centreline for each of the mesh grid base sizes considered. Based on the Mach contours supplied by Mehta *et al.* (See Figure 10a), three shock cells are expected, and from the reviewed literature pertaining to this flow phenomenon [23] [22], it is understood that a strong deceleration should be present and discontinuous across the terminus of each of the shock cells. Although this general behaviour was reproduced in each simulation, the discontinuous deceleration was best captured by the simulation evaluated for a base size of 0.02 m. The smaller base size of 0.01 m yielded unsteady flow velocity, and the larger base size of 0.03 m did not adequately capture the discontinuous flow, indicated by the spurious data points between the plotted curves. To conclude, the simulation computed using a base size of 0.02 m was the most optimal, and only its results were carried forward for further analysis.

4.2.2 Simulation Results

Available overleaf are the Mach contours obtained for the 2-D quarter-scale CFD simulation of the Mars Space Laboratory (MSL) Mars Landing Engine (MLE) nozzle, reproduced from data supplied in *Thruster Plume Surface Interactions: Applications for Spacecraft Landings on Planetary Bodies* by Mehta *et al.*, 2013 (available in Reference [16]). Note that Mach contours have been mirrored about the nozzle axis.

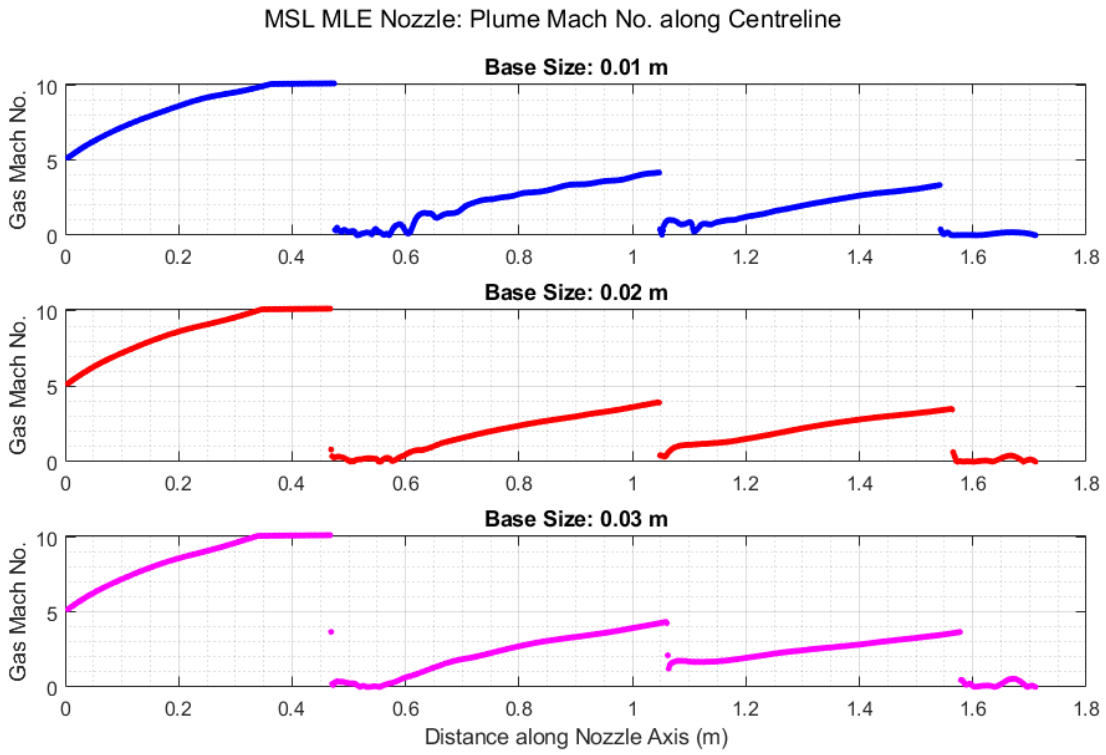


Figure 28: 2D MSL MLE Nozzle Simulation: Sensitivity Analysis, Mach No. along Nozzle Axis

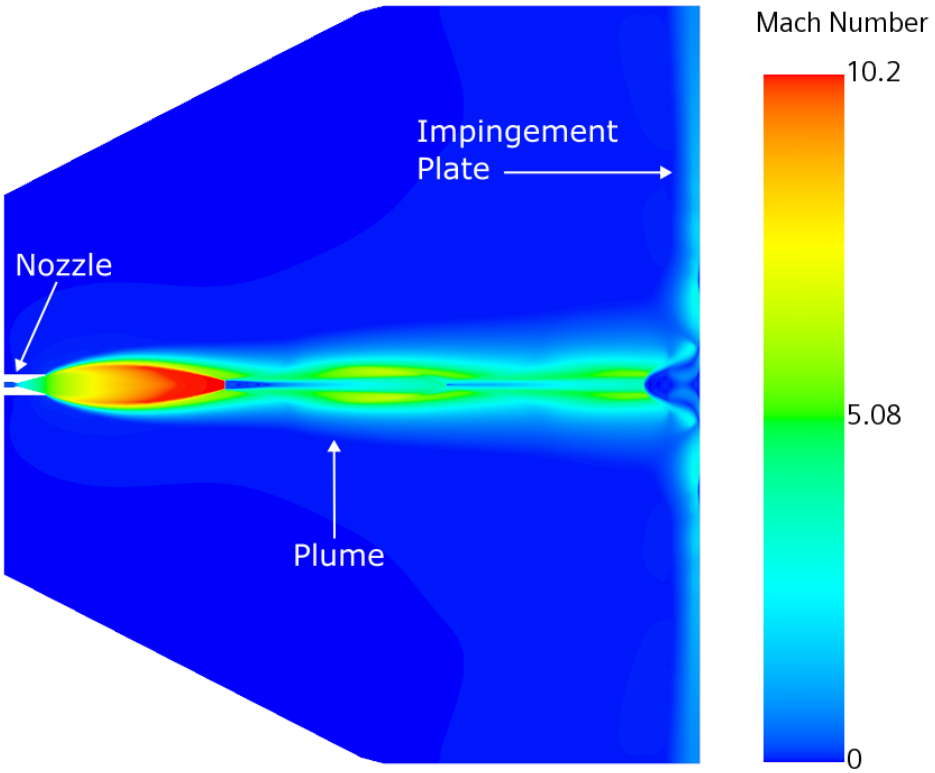


Figure 29: Reproduced Quarter-Scale MSL MLE Nozzle Simulation, Mach Contours

4.2.3 Analysis and Validation

The reproduced quarter-scale MSL MLE nozzle Mach contours given in Figure 29 are rotated in sense by 90° anticlockwise compared to the Mach contours published by Mehta *et al.*, available in Figure 10a, due to the spatial modelling strategy employed. However, both plume flows are superficially similar, including three distinct shock cells, recognisable from surveyed literature on under-expanded flow structures [23], with the characteristic cyclic contraction and expansion profile and corresponding flow acceleration. The nozzle exit Mach number of the simulated flow was sampled and evaluated as approximately Mach 5.1, evidenced in Figure 30 below, which is very similar to the value prescribed by Mehta *et al.* of Mach 5.0.

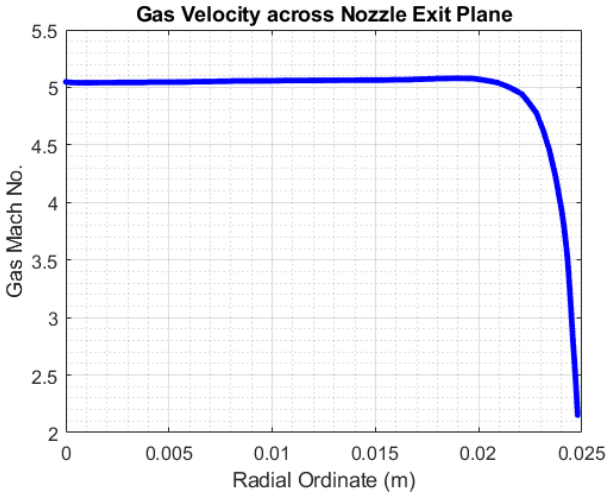


Figure 30: Reproduced Quarter-Scale MSL MLE Nozzle Simulation, Nozzle Exit Mach No.

The reproduced MSL MLE simulation incorporated viscous flow physics models with applied viscous forces, responsible for the curvature of the nozzle exit plane velocity profile towards the nozzle wall located at a radial ordinate of 24.79 mm from the nozzle axis. The characteristic shape depicted in Figure 30 indicates the action of simulated viscous forces, which reduced the gas velocity to zero at the nozzle's internal surface. Of interest is the strong Mach disc captured at the terminus of the first shock cell and the weaker Mach disc captured at the second. Literature regarding this phenomenon [22] concludes that Mach discs should indeed exist in this flow, in part owing to the large pressure ratio of the MSL MLE nozzle. Nevertheless, no Mach discs are present in the results published by Mehta *et al.* which may suggest that the reproduced simulation was of comparatively greater resolution and generated flow effects in finer detail, a quality afforded by the reduced size of the simulation domain of the reproduced work.

An obvious divergence from the published Mach contours is the structure of the reproduced flow at the impingement plate, notably the highly-curved bow shock rather than the standing shock wave seen in the original work. The presence of a curved bow shock giving rise to a bubble of stagnated flow is, however, in keeping with current knowledge of PSI phenomenology [25], and similar flow structures

were observed by Lamont and Hunt [26], hence the obtained results were not considered incorrect. It is likely that the discrepancy is the result of differing steady and unsteady physics models between the original and reproduced simulations, generating dissimilar time-averaged 'images' of oscillating [26] shock waves at the impingement point. Evident too is the recirculation of flow within the stagnation bubble disclosed by the curved regions of non-zero Mach number flow inside the bubble, another captured flow behaviour documented in published literature [16]. A final observation of the reproduced Mach contours is the presence of a triple point and tail shock wave leading to radially-propagating wall flow within a clearly-defined shear layer, a similarity shared with both the original work and the studied literature on impinging flows [25]. For clarity, each of the aforementioned flow features are indicated below in Figure 31, which depicts a compound image of the reproduced Mach contours and an algorithmically-obtained pseudo-Schlieren image to illuminate the various supersonic flow structures. The numerical Schlieren image was synthesised using a series of scalar field functions applied across the simulated domain extent to identify strong changes in gas density, where a bright area of the image indicates a strong change in density emblematic of a shock wave or compression wave.

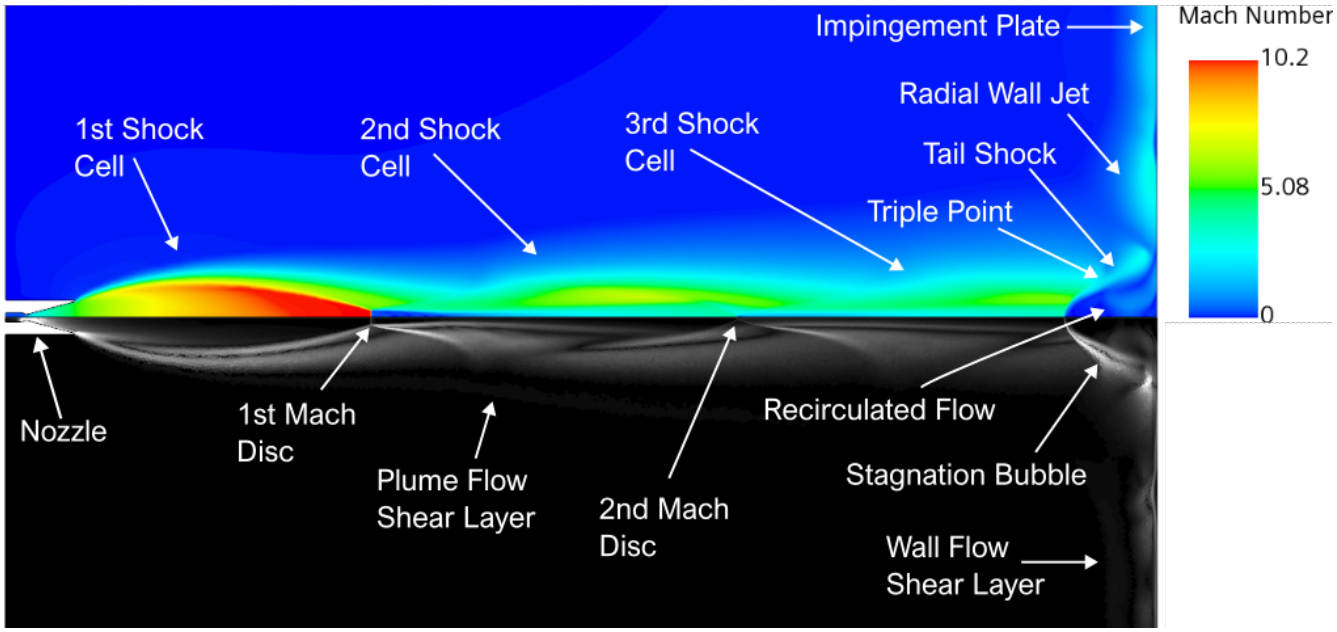


Figure 31: Reproduced Quarter-Scale MSL MLE Nozzle Simulation, Schlieren Image and Mach Contours

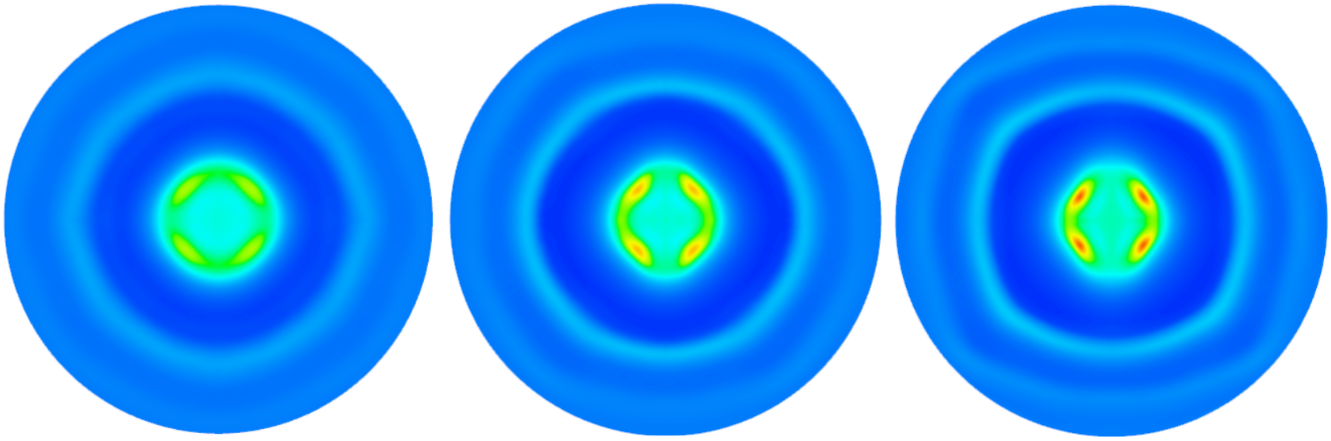
Given the developed and equipped numerical simulation method correctly reproduced the majority of the published Mach contours that informed its development (see Figure 10a), and even captured additional flow effects beyond those in the original work where the few differences present were found to be literature-accurate, the numerical method was deemed capable of successfully simulating correct and accurate supersonic plume and impinging flows.

4.3 3D Supersonic Fluidic Injector Simulations

The results obtained for the *injector off*, *injector on*, and *canted injector* simulations are presented in unison to better illustrate the favourable flow conditions imparted by perpendicular supersonic injector flow, and the further improvements made to the impinging flow by adjusting the injector line-of-thrust to introduce a gas swirl velocity.

4.3.1 Mesh Base Size Sensitivity Analysis

The 3-D supersonic fluidic injector simulation with no flow prescribed through the injector nozzle was evaluated for base sizes of 7 mm, 9 mm, and 11 mm to probe whether the choice of mesh grid base size significantly influenced the results obtained. The focus of this analysis was on the resolution in which each simulation captured the plume-induced distribution of pressure across the simulated impingement plate, as surface pressures are a core feature of plume-surface effects [4]. The use of the MSL MLE nozzle not only removed a point of uncertainty as this nozzle behaviour was now well understood, but the impingement pressure profile published by Mehta *et al.* for this nozzle, reproduced in Figures 11a and 11b in Section 2.3.2, provided a guide for the general arrangement of surface pressure likely to be obtained for the injector simulations. The lack of direct applicability of the published pressure footprint arose from the impingement altitude of 34.5 d considered by Mehta *et al.* versus the 10 d used for the present supersonic injector simulations.



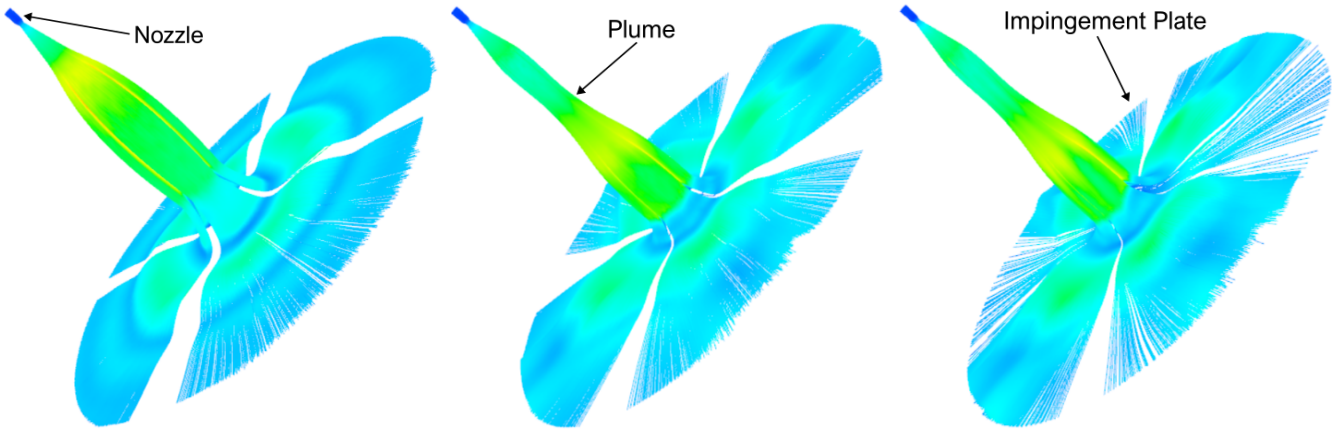
(a) Base Size: 7 mm (b) Base Size: 9 mm (c) Base Size: 11 mm

(d) Pressure Magnitude and Normalised Pressure Colour Bar

Figure 32: Supersonic Fluidic Injector Sensitivity Analysis: Impingement Pressure Distribution

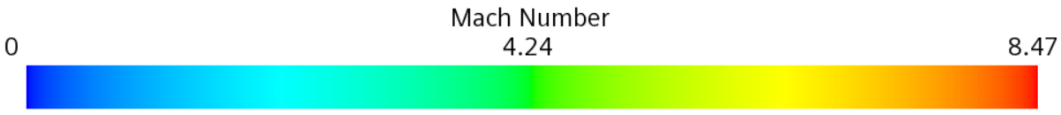
Presented in Figures 32a to 32c previous are the impingement pressure distributions obtained for mesh grid base sizes of 7 mm, 9 mm, and 11 mm respectively, where the full 360° footprint was produced by mirroring the quarter-domain about its symmetry axes. A combined colour bar of pressure magnitude and impingement pressure, P_g, normalised by nozzle chamber pressure, P_c is included for reference in Figure 32d. Reviewing Figures 32, immediately evident is the visual disparity between the obtained pressure distributions and the pressure profile published by Mehta *et al.* for the same nozzle (available in Figure 11a), as well as the approximately halved normalised impingement pressure for the obtained pressure distributions despite the 60% reduction in impingement altitude. This was likely the result of the 22.5° angle of inclination of the MSL MLE nozzle in the simulations generated by Mehta *et al.*, but further testing would be required to strengthen this hypothesis. Nevertheless, the pressure footprint for a mesh grid base size of 9 mm, available in Figure 32b, appears to have included the pressure profile in the greatest fidelity of the three base sizes considered. This is evidenced by the well-defined rings of comparatively large pressure, a result of the arrangement of radially propagating annular wall jets. For clarity, this flow phenomenon is captured in Figure 31 in the section previous, and illustrated in Figure 5a in Section 2.1.5. The same rings are less sharp in Figure 32b and are of nonphysical shape in Figure 32c. The small regions of high pressure toward the centre of each pressure distribution suggest a minor incompatibility of this flow with the exploitation of symmetry, as such 'hot-spots' would not exist for uniform flows and would instead present as complete rings [16], if at all. To conclude, the intermediate mesh grid base size of 9 mm struck a balance between simulation accuracy and computational economy and proved most optimal, thus was used henceforth for the supersonic fluidic injector simulations.

4.3.2 Simulation Results



(a) No Injector Flow (b) Perpendicular Injector Flow (c) Canted Injector Flow

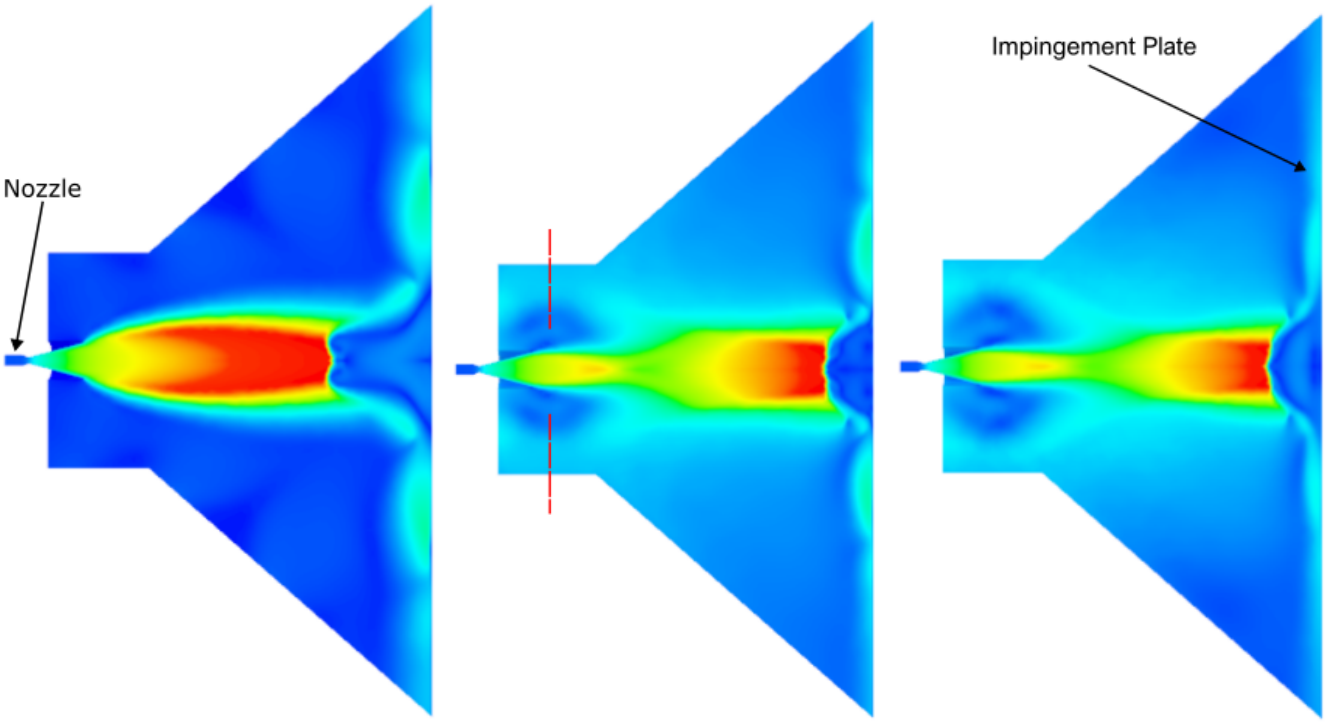
Figure 33: Supersonic Fluidic Injector Simulation: Streamsurfaces for Each Injector Configuration



(d) Streamsurface Colour Bar: Gas Mach Number

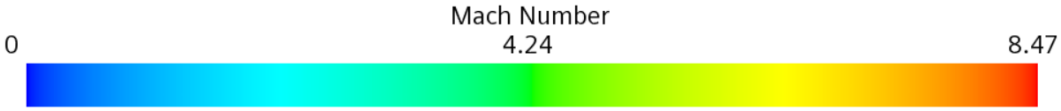
Figure 33: Supersonic Fluidic Injector Simulation: Streamsurfaces for Each Injector Configuration

Given in Figures 33a to 33c previous are the visualisations of flow streamlines in 3-D space, known as a *streamsurface*, obtained for each of the three injector configurations simulated. Namely, these are no injector flow, perfectly perpendicular injector flow, and canted injector flow respectively. These streamsurfaces were used to provide a visual aid in qualitatively understanding how supersonic fluidic injector technology modified the general shape of an impinging flow. A colour bar of gas Mach number is included for reference in Figure 33d. Exploiting cyclic symmetry, the simulated 90°-revolve domain was transformed to produce a full 360° revolution. To more quantitatively monitor the flow manipulation achieved using supersonic injector technology, Mach contours were generated for the entire domain under the same three injector conditions and are available below in Figures 34a to 34c. The red dashed lines in Figure 34b indicate the location and thrust lines of the two injectors present in the 180° revolve of the domain. A colour bar of gas Mach number is available overleaf in Figure 34d for reference.



(a) No Injector Flow (b) Perpendicular Injector Flow (c) Canted Injector Flow

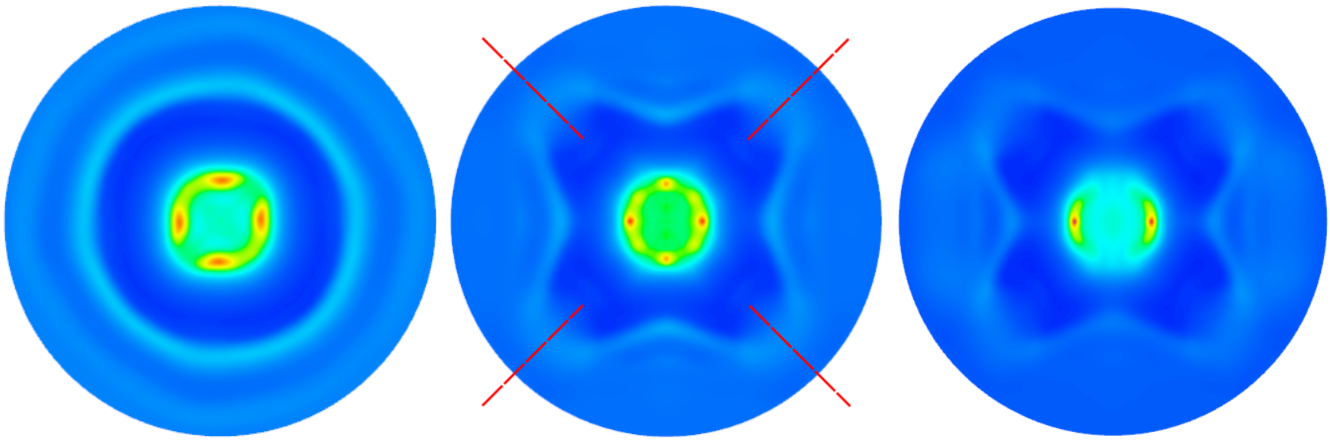
Figure 34: Supersonic Injector Simulation: Plume Mach Contours for Each Injector Configuration



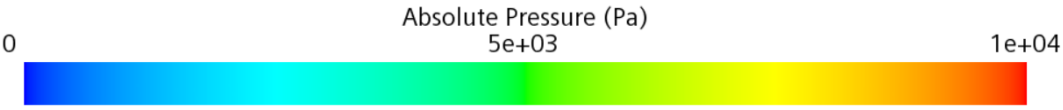
(d) Mach Contours Colour Bar: Gas Mach Number

Figure 34: Supersonic Injector Simulation: Plume Mach Contours for Each Injector Configuration

Cyclic symmetry was again leveraged to produce the Mach contours for a 180°-revolve. Lastly, to explore the influence of the supersonic injector-induced jet-jet interaction on the distribution of surface pressure exerted by the plume on the impingement plate, referred to as impingement pressure, pressure contours were generated across the simulated impingement plate. The profile of impingement pressure for a given nozzle flow is referred to as the pressure *footprint*. Contours were synthesised for the injector states of no injector flow, perpendicular injector flow, and canted injector flow, and are given below in Figures 35a to 35c, including a colour bar of absolute pressure in Figure 35d. Once again, the red dashed lines in Figure 35b indicate the radial locations of the four injectors included in a 360° revolve of the simulation domain.



(a) No Injector Flow (b) Perpendicular Injector Flow (c) Canted Injector Flow



(d) Pressure Distribution Colour Bar: Absolute Pressure

Figure 35: Supersonic Fluidic Injector Simulation: Impingement Pressure Profile for Each Injector Configuration

4.3.3 Analysis

Inspection of the streamsurfaces shown in Figures 33 reveals that the approximate action of the supersonic injector-induced jet-jet interaction was to narrow the primary nozzle plume at the point where it inter-

sected with the injector plumes, which further downstream resulted in a smaller zone of recirculated flow. Note that the flow from the injector plumes was omitted for clarity, and the gaps between streamlines were the consequence of simulating only a quarter of the domain and employing geometric symmetry to retroactively construct the full domain. Streamlines were initialised at the nozzle inlet surface. Consulting the colour bar given in Figure 33d, it is evident that both the perpendicular and canted jet-jet interactions depicted in Figures 33b and 33c respectively induced an approximately constant gas velocity in the primary plume local to its intersection with the injector plumes, delaying the onset of flow acceleration to a point further downstream. Most critical is the flattening of the 'disc' of radially-propagating flow at the impingement plate conspicuous in both figures, illustrating a reduction in the redirection of the wall flow due to weakened wall jets. The diffused changes in colour demonstrate reduced flow velocity gradients. Canting the injectors marginally improved the flattening and distributing of the radially-propagating flow, shown in Figure 33c. The Mach contours for a half-domain view presented in Figures 34 provide a more quantitative overview of the favourable flow manipulation achieved using supersonic fluidic injector technology, but naturally exhibit identical behaviours. Comparing Figures 34b and 34c to Figure 34a, the stagnation bubble was reduced in dimension for the plume flows simulated with both the perpendicular and canted species of jet-jet interaction. The radial wall jets and tail shock were similarly reduced in size and intensity by the perpendicular jet-jet interaction, given in Figure 34b, and were further reduced for the case of canted injector flow in Figure 34c where redirection of wall jet flow was negligible and radial flow dissipated rapidly. Interestingly, the expansion and subsequent compression of gas flow along the plate, indicated by the flow becoming turned into the domain at the site of the wall jets, was greatly reduced by the application of perpendicular injector flow and all but removed for canted injector flow. The application of supersonic fluidic injector technology improved the uniformity of gas flow across the impingement surface, and the plume gas Mach number was notably reduced for the majority of the expanding plume flow by the injectors.

Directing attention to the application of surface pressure by the impinging plume, unequivocally the most significant and decisive flow feature, the profiles of pressure distribution depicted are in Figures 35 for the conditions of no injector flow, perpendicular injector flow, and canted injector flow respectively. These figures illuminate the diffused impingement pressure achieved by both the perpendicular and canted jet-jet interactions, given in Figures 35b and 35c respectively, with the injector-off condition displayed in Figure 35a. The ring shape exhibited for the simulation without injector flow arose from the presence of radial wall jets in a planar sense, visualised here as an annular wall jet, which was distorted by the introduction of injector flow. In detail, the application of perpendicular injector flow dissipated much of the comparatively high-pressure annular wall jet and reduced the pressure gradients local to it, disclosed by the diffused colour representations in Figure 35b. Canted injector flow achieved the same result to

a greater degree, given in Figure 35c showing the further dissipated radial flow. As observed for the streamsurfaces and Mach contours, the injector-induced jet-jet interactions reduced the diameter of the stagnation bubble, illustrated as the circular zone of high pressure in the centre of each footprint. An additional and unexpected observation can be made that the surface pressure induced by the stagnation bubble was increased by the introduction of perpendicular injector flow, displayed by the shift in colour toward the maximum pressure (red) end of the colour bar spectrum, yet was reduced for the application of canted injector flow, revealed by a shift in colour towards the zero-pressure (blue) end of the colour bar spectrum. Further and final study of the applied surface pressures for each simulated injector condition was facilitated by sampling the gas pressure at the surface of the impingement plate along a path parallel to the injector line-of-thrust, demarcated by the red dashed lines in Figures 34b and 35b. These data were complemented by surface shear stress probed along the same path, as surface pressure and shear stresses were understood to be the driving forces behind most modes of plume-driven surface alteration [16] [25] [26]. Both data sets are available below in Figures 37 and 36, including a legend to distinguish each curve.

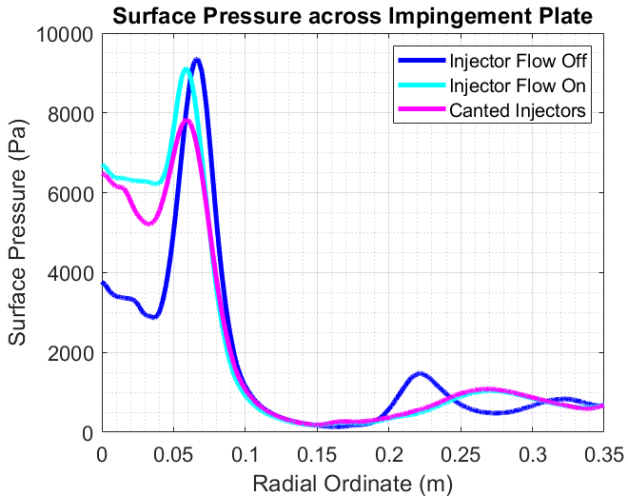


Figure 36: Supersonic Fluidic Injector Simulation: Impingement Pressure versus Distance from Plume Axis

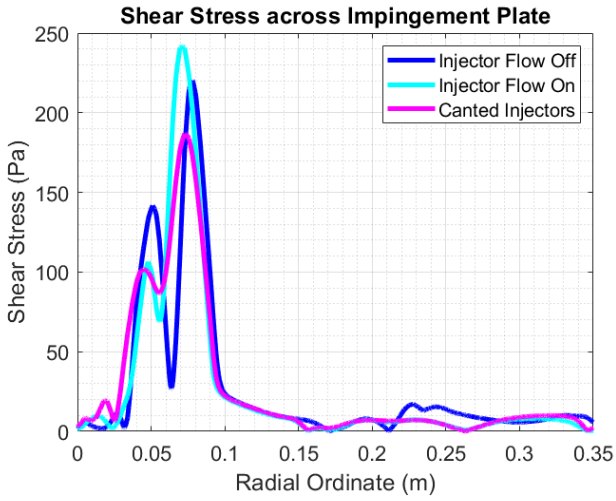


Figure 37: Supersonic Fluidic Injector Simulation: Shear Stress on Impingement Plate versus Distance from Plume Axis

From inspection of the plot of surface impingement pressures for the injector states of no flow, perpendicular flow and canted flow, given in Figure 36, it is clear that the maximum impingement pressure was reduced and the pressure distribution smoothed by perpendicular jet-jet interaction, where the plotted peaks correspond to the planar wall jets presented in Figures 34 and the annular wall jet illustrated in Figures 35. The maximum surface pressure was further reduced for canted supersonic fluidic injector flow. Unexpectedly, for both the perpendicular and canted injector simulations the pressure at the centre

of the stagnation bubble, corresponding to a radial ordinate of 0 m, was increased beyond that for the injector-off condition, aligning with the same observation for the pressure profiles analysed in Figures 35. Pronounced in Figure 37, the plot of surface shear stress versus radial ordinate across the impingement plate for each simulated injector state, is that whilst the introduction of perpendicular injector flow increased the maximum shear stress across the plate when compared to the plotted data for no injector flow, the introduction of canted injector flow reduced the surface shear stress across the sampled path and also smoothed the shear stress distribution. In numbers, the application of canted supersonic fluidic injector technology to the simulated impinging flow yielded a reduction in maximum impingement pressure of almost 2000 Pa in combination with a reduction in maximum shear stress of 60 Pa. These values correspond to reductions from maximum of approximately 21% and 27% respectively. Finally, to confirm the presence of the swirl-induced artificial pressure field the canted supersonic fluidic injectors were hoped to yield, gas pressure was sampled within a region of the expanded plume an arbitrary distance above the impingement plate on a path parallel with the injector line-of-thrust. The obtained data are given below in Figure 38, where data were sampled within the space between the outer edge of the plume core and the inner edge of the domain extent to avoid corrupting the measurement with the large gas pressure invariably present in any plume core. Immediately evident in the figure is that a lower gas pressure was present closer to the centre of the plume (yet outwith the plume core) than at its perimeter, indeed providing the intended artificial pressure field which acted towards the primary plume centreline or nozzle axis.

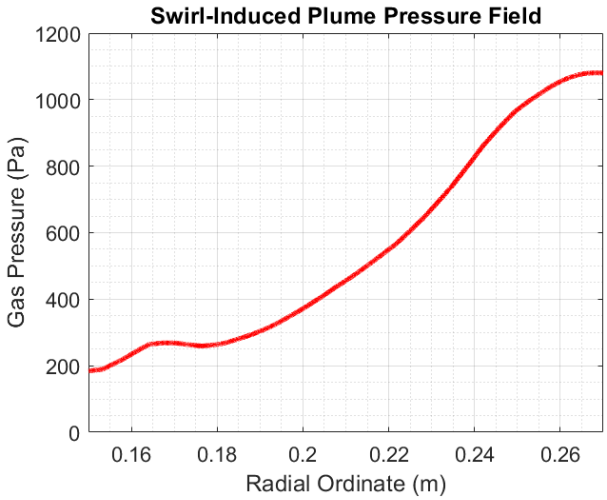


Figure 38: Supersonic Fluidic Injector Simulation: Artificial Pressure Field

In summary, the application of a supersonic jet-jet interaction to a simulated impinging plume flow by means of supersonic fluidic injectors was observed to distribute flow velocity and pressure more evenly across the impingement plate, diffuse concentrated regions of comparatively large surface pressure, reduce

the strength and redirection of radial wall jets propagating across the impingement surface, reduce the maximum impingement pressure imparted by the expanding plume, and reduce the height and diameter of the stagnation bubble as well as the surface pressure it imparts. Adjusting the arrangement of these injectors to introduce a gas swirl velocity produced these positive flow modifications to a greater extent, yielding still-improved distribution and smoothing of both radial gas velocity and impingement pressures across the impingement surface, coupled with further reduced maximum impingement pressure that was evenly applied. Unique to the canted injector interaction was a reduction in the maximum shear stress on the impingement plate coupled with radial wall flow that was significantly reduced in strength. These flow modifications are extremely beneficial to reducing the negative properties of PSI, as these are driven principally by focused impingement pressures and large pressure gradients [4] [16], as well as shock wave - surface interactions effected by stagnated and recirculated flow [12]. Numerous operational challenges during powered landings stem from the propagation of radial flow outward from the point of impingement [2], a flow characteristic minimised by the application of canted supersonic fluidic injectors. A final property of the artificially-swirled flow was the verified presence of the desired pressure field which acted towards the primary plume centreline or nozzle axis, intended to decelerate and capture regolith ejecta.

4.3.4 Validation

Confirming the validity of the results obtained for the supersonic fluidic injector simulations proved challenging, as limited published works documenting the use of this technology exist. Kobayashi and Ukai described a total elimination of shock-cell structures beyond the point of the induced jet-jet interaction [9], and an attempt was made to produce pseudo-Schlieren images of density change across the entire domain to investigate whether the obtained results demonstrated this behaviour. However, the size of the domain required that the element size across which local density change was computed was exceedingly large to negate extreme computational expense, and therefore produced images of unusable quality. To demonstrate and for completeness, these images are included in Appendix B. Whilst the gas compression due to the jet-jet interaction was captured, it can be seen that no shock cell structures were captured despite the certainty [16] [22] that they must exist in the flow, which was considered to be the result of an unsuitably large element size. Nonetheless, Kobayashi and Ukai demonstrate a radially-expanded flow as a result of their supersonic jet-jet interaction technique, which was reflected in the results obtained in this study, and also document a reduction in maximum surface stagnation pressure of 48%, which suggests that the reduction in maximum impingement pressure achieved for the canted injector flow of around 21% was not unrealistic.

Reviewing Figures 34 for the 3-D supersonic injector simulation and Figure 29 for the 2D MSL MLE nozzle simulation, flow through both the 2-D and 3-D geometries of the nozzle used to represent the

MSL MLE nozzle behaved in identical manners. Both flows exhibited even flow acceleration through the nozzle, disclosed by the uniform gradient of colour as the flow velocity increased, in tandem with a uniform nozzle exit velocity of around Mach 5 understood by consulting their respective colour bars. This is an encouraging observation, and the obtainment of similar flow properties for the same nozzle geometry using different spatial modelling approaches and simulation conditions strengthens the results produced from both simulations. Similarly, the general arrangement of the flow around the stagnation bubble is visually similar between both the reproduced MSL MLE nozzle flow and the supersonic injector simulation, further assuaging the initial reservations held when analysing the results of the reproduced MSL MLE simulation that this flow was inaccurate. Additionally, the pressure footprint generated for the MSL MLE nozzle plume flow by Mehta *et al.*, reproduced in Figure 11a in Section 2.3.2, depicts a maximum impingement pressure much greater than that achieved for the simulated *injector off* condition, seen in Figures 35a and 36; this is likely due to the angle of incidence of the MSL MLE nozzle simulated by Mehta *et al.* focusing impingement pressures. The smaller obtained value suggests that the pressure profiles developed for the injector simulations were within the boundaries of physical values. Although further validation is certainly desirable, these properties, coupled with the generation of the supersonic fluidic injector simulations using a proven and validated numerical simulation method, allow the obtained results to be considered valid with a good degree of confidence.

5 Discussion

5.1 McCabe Nozzle Flow Simulation

The results obtained for the McCabe nozzle flow simulation were of good quality and literature-accurate, and were wholly satisfactory. The verified and high-quality output of this simulation proved that the developed method of numerically simulating 2-D nozzle flow in the near-field regime was valid and accurate.

5.2 MSL MLE Nozzle Simulation

For the MSL MLE nozzle simulation, whilst the results were ultimately deemed valid and also literature-accurate, it would be desirable to execute the same simulations for various nozzle profiles to investigate whether the unusual impinging flow and stagnation bubble arrangement were present for all geometries. Nonetheless, the reproduced flow to represent the exhaust plume of the *SkyCrane* vehicle matched the published work used to build the simulation method satisfactorily and even captured flow effects not included in the published work. The verified output of this simulation, too, proved that the numerical simulation methodology used to produce it was valid.

5.3 Supersonic Fluidic Injector Simulation

Considering the supersonic fluidic injector simulations, it was desirable to compute a greatly increased number of streamlines to synthesise a more complete streamsurface, however, this proved prohibitively computationally expensive. Similarly, the 'hot-spots' present in the provided Mach contours indicate the requirement for a finer mesh, and the domain-filling extent of the flows detailed in Figures 34b and 34c illuminate the need for a larger domain to ensure all flow effects are accurately represented. The barrier to both refining the mesh grid and enlarging the domain was computational cost. For clarity, all simulations pertaining to supersonic fluidic injector technology were executed in parallel across up to 48 physical cores using the University of Glasgow's High-Performance Computing (HPC) cluster, yet even greater computational power may be required to produce Computational Fluid Dynamics (CFD) simulations of hypersonic impinging flows using advanced turbulence schemes and gas flow models in greater resolution than was done in this work.

6 Final Conclusions

The first aim of developing a valid and reliable method of numerically simulating hypersonic impinging plume flows according to current knowledge of Plume Surface Interactions (PSI) phenomenology was successfully achieved, as the developed simulation method was proven to accurately model nozzle flow in the near-field regime and impinging flow in the far-field regime. The applied numerical methodology generated valid flow through a 2-D boundary-layer-corrected supersonic nozzle as well as impinging flow to represent plume-surface effects encountered during the low altitude hovering of the *SkyCrane* vehicle, both of which were published works used to inform the development of the method. Whilst the scope for improvement was identified, the simulation method was trusted to produce valid output with a good degree of confidence.

The second aim of theorising and undertaking the preliminary evaluation of a novel and applicable method of controlling PSI using only spacecraft-borne hardware was also successfully achieved. The application of canted supersonic fluidic injector technology was observed to reduce the maximum impingement pressure and surface shear stress induced by the plume, smooth the distribution of pressure and radial flow velocity across the impingement plate, dissipate the annular wall jet and succeeding radially-propagating plume flow, reduce the diameter of the stagnation bubble and the magnitude of the surface pressure it applied, and induce an artificial pressure field designed to decelerate and contain regolith ejecta. It is worth reiterating that the theorised mitigation strategy of leveraging secondary flow to contain regolith ejecta coupled with equipping supersonic fluidic injector technology to minimise the extreme shock-surface inter-

actions that occur during impingement is not intended to be a complete solution. Instead, it is likely that a successful and complete method of reducing destructive plume-surface effects such that repeated and safe planetary landings can be executed would involve the use of a spacecraft-based mitigation method, such as the one theorised and evaluated in this study, applied in tandem with a surface-based mitigation method of high Technology Readiness Level (TRL), such as regolith sintering or the injection of landing engine plumes with alumina. Regardless, there are scenarios in which the application of supersonic fluidic injector technology presented in this work could be used individually, such as to facilitate the initial safe transport of equipment and engineering hardware to construct surface infrastructure on a planetary body to enable subsequent use of combined spacecraft-based and surface-base methods of controlling plume-surface effects for future missions.

7 Proposed Future Work

7.1 Simulation Methodology

It would be desirable to run the MSL MLE simulation in 3-D space so that the effect of adjusting the nozzle's line of thrust to match the true *SkyCrane* configuration could be investigated. It would also be beneficial to substitute the conical supersonic nozzle geometry employed in place of the real MSL MLE nozzle geometry for various other nozzle profiles to further explore the unusual configuration of the flow around the stagnation bubble.

7.2 Canted Supersonic Fluidic Injectors

The application of supersonic fluidic injector technology to control destructive plume-surface effects presented in this work is considered worthy of further study, as preliminary evaluation shows it has real value in controlling PSI such that safe landings of spacecraft with large dry mass are possible, and could be realised with minimal design adaptations to existing spacecraft landing engines and nozzle configurations. A significant flaw in this work was the omission of simulations designed to monitor the transient properties of canted injector flow, done only to avoid encountering greater computational expense. It is proposed that topics for future study of this technology include the time-variation of impingement pressures as the plume expands, and also as the spacecraft descends. Further topics that would be useful to the analysis of canted jet-jet interactions include an in-depth review of the effect of various injector incidence angles, size, design Mach number, and location relative to the primary nozzle exit plane on both surface impingement pressure and surface shear stress. The nozzle performance after a certain amount of flow is siphoned for use with the supersonic injectors should be explored, as well as the possibility of using cold gas instead.

References

[1] M. Kuhns, P. Metzger, A. Dove, and J. Byron, "Deep regolith cratering and plume effects modeling for lunar landing sites," *Earth and Space Conference*, 2021.

[2] P. T. Metzger, J. Smith, and J. E. Lane, "Phenomenology of soil erosion due to rocket exhaust on the moon and the mauna kea lunar test site," *Journal of Geophysical Research*, 2011.

[3] P. T. Metzger, J. E. Lane, C. D. Immer, *et al.*, "Scaling of erosion rate in subsonic jet experiments and apollo lunar module landings," 2008.

[4] R. N. Watkins, M. Mehta, D. Han, P. Prem, and L. Sibille, "Understanding and mitigating plume effects during powered descents on the moon and mars," 2021.

[5] M. Mehta, N. O. Renno, J. Marshall, *et al.*, "Explosive erosion during the phoenix landing exposes subsurface water on mars," *Icarus*, 2011.

[6] D. Budzyń, H. Zare-Behtash, A. Cowley, and A. Cammarano, "Implicit lunar dust mitigation technology: Compliant mechanisms," *Acta Astronautica*, vol. 203, pp. 146–156, 2023.

[7] A. Che, R. Zhu, and M. Jiang, "Dynamic response of lunar soil caused by landing impact," *KSCE Journal of Civil Engineering*, 2018.

[8] A. Einstein, "Die ursache der meanderbildung der flusslaufe und des sogenannten baerschen gesetzes," *Die Naturwissenschaften*, 1926.

[9] M. Kobayashi and T. Ukai, "Effectiveness of a supersonic jet-jet interaction technique on flow structure in an underexpanded jet impinging on a perpendicular plate," *Acta Astronautica, Elsevier*, 2022.

[10] R. P. Mueller, N. J. Gelino, K. L. Dixon, L. Sibille, and B. T. Vu, "Large vehicle lunar landing surface interaction and in-situ resource based risk mitigation: Landing & launch pads,"

[11] L. S. I. Consortium, "Lsic spring meeting program," 2021.

[12] M. Mehta, "Rocket plume interactions for nasa landing systems," *NASA - DLR Technical Interchange Meeting*, 2019.

[13] J. Park, Y. Liu, K. D. Kihm, and L. A. Taylor, "Characterization of lunar dust for toxicological studies. i: Particle size distribution," *Journal of Aerospace Engineering, ASCE*, 2008.

[14] J. E. Lane, P. T. Metzger, and C. D. Immer, "Lagrangian trajectory modeling of lunar dust particles," *ASRC Aerospace, NASA/KSC Granular Mechanics and Surface Systems Laboratory*, 2008.

[15] R. N. Clegg, B. L. Jolliff, M. Robinson, B. W. Hapke, and J. Plesica, "Effects of rocket exhaust on lunar soil reflectance properties," *Icarus*, 2014.

[16] M. Mehta, A. Sengupta, N. O. Renno, and J. W. V. Norman, "Thruster plume surface interactions: Applications for spacecraft landings on planetary bodies," *American Institute of Aeronautics and Astronautics*, 2013.

[17] M. T. Lemmon, S. D. Guzewich, T. G. McConnochie, *et al.*, "Martian dust particle size during the 2018 planet-encircling dust storm as measured by the curiosity mars rover," *Ninth International Conference on Mars*, 2019.

[18] C. Immer, P. T. Metzger, P. E. Hintze, A. Nick, and R. Horan, "Apollo 12 lunar module exhaust plume impingement on lunar surveyor iii," *Icarus*, vol. 211, 2011.

[19] C. Immer, J. Lane, P. Metzger, and S. Clements, "Apollo video photogrammetry estimation of plume impingement effects," *Icarus*, 2011.

[20] J. Gómez-Elvira, C. Armiens, I. Carrasco, *et al.*, "Curiosity's rover environmental monitoring station: Overview of the first 100 sols," *Journal of Geophysical Research: Planets*, vol. 119, no. 7, 2014.

[21] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 7th ed. New York: Wiley, 2001.

[22] D. Edgington-Mitchell, D. R. Honnery, and J. Soria, "The underexpanded jet mach disk and its associated shear layer," *Physics of Fluids*, 2014.

[23] J. A. Inman, P. M. Danehy, R. J. Nowak, and D. W. Alderfer, "Fluorescence imaging study of impinging underexpanded jets," *46th AIAA Aerospace Sciences Meeting and Exhibit*, 2008.

[24] A. Michael, "Mach disk in underexpanded exhaust plumes," *AIAA Journal*, vol. 9, no. 3, pp. 512–514, 1971.

[25] J. Pattison, S. Celotto, R. Morgan, M. Bray, and W. O'Neill., "Cold gas dynamic manufacturing: A non-thermal approach to freeform fabrication," *International Journal of Machine Tools and Manufacture*, vol. 47, no. 3, pp. 627–634, 2007.

[26] P. J. Lamont and B. L. Hunt, "The impingement of underexpanded axisymetrical jets on perpendicular and inclined flat plates," *Journal of Fluid Mechanics*, vol. 100, 1980.

[27] K. W. Farries, P. Visintin, S. T. Smith, and P. van Eyk, "Sintered or melted regolith for lunar construction: State-of-the-art review and future research directions," *Construction and Building Materials*, 2021.

[28] M. Kuhns, P. Metzger, Z. Hasnain, and K. Zacny, "Instant landing pads for lunar missions," *Earth and Space Conference*, 2021.

[29] A. McCabe, "Design of a supersonic nozzle," 1967.

[30] D. K. Huzel and D. H. Huang, *Modern Engineering Design of Liquid Propellant Rocket Engines* (Rocektdyne Division, North American Aviation, Inc.). Scientific and Technical Information Division, Office of Technology Utilisation, NASA, 1966.

[31] H. Schlichting and K. Gersten, *Boundary Layer Theory*, 8th ed. Springer-Verlag Berlin Heidelberg, 2000.

[32] T.Cebeci, J. P. Shao, F. Kafyeke, and E. Laurendeau, *Computational Fluid Dynamics for Engineers*. Horizons Publishing, Springer, 2005.

[33] K. Davis, E. Fortner, M. Heard, H. McCallum, and H. Putzke, "Experimental and computational investigation of a dual-bell nozzle," 2014.

[34] J. A. Pohly, C.-k. Kang, D. B. Landrum, J. E. Bluman, and H. Aono, "Data-driven cfd scaling of bioinspired mars flight vehicles for hover," *Acta Astronautica*, 2020.

[35] G. V. R. Rao, "Exhaust nozzle contour for optimum thrust," *Journal of Jet Propulsion*, 1958.

A Tabulated McCabe Nozzle Co-Ordinates

Table 12: Boundary-Layer-Corrected Nozzle Co-Ordinates. Adapted from Reference [29]

x'	y'	x'	y'	x'	y'
0.0000	2.6360	5.5000	0.5924	8.0561	1.2395
0.5000	2.4097	5.5957	0.5962	8.3488	1.3294
1.0000	2.1848	5.6823	0.6033	8.5615	1.3928
1.5000	1.9168	5.7812	0.6122	8.9019	1.4897
2.0000	1.7414	5.9096	0.6289	9.2759	1.5893
2.5000	1.5248	6.0159	0.6478	9.6865	1.6911
3.0000	1.3138	6.1438	0.6747	10.1373	1.7946
3.5000	1.1117	6.2892	0.7085	10.6322	1.8992
4.0000	0.9241	6.4291	0.7431	11.1753	2.0042
4.5000	0.7616	6.5755	0.7836	11.7716	2.1088
4.6000	0.7336	6.6636	0.8082	12.4262	2.2119
4.7000	0.7075	6.8324	0.8574	13.1447	2.3123
4.8000	0.6835	6.9493	0.8924	13.9339	2.4086
4.9000	0.6619	7.0804	0.9330	14.8006	2.4992
5.0000	0.6428	7.3283	1.0104	15.7529	2.5819
5.1000	0.6265	7.5008	1.0645	16.7996	2.6545
5.2000	0.6131	7.5807	1.0897	17.9504	2.7142
5.3000	0.6029	7.7265	1.1357	19.2166	2.7577
5.4000	0.5960	7.8414	1.1719	20.0316	2.7759

All dimensions in inches

B Supersonic Fluidic Injector Simulation: Pseudo-Schlieren Images

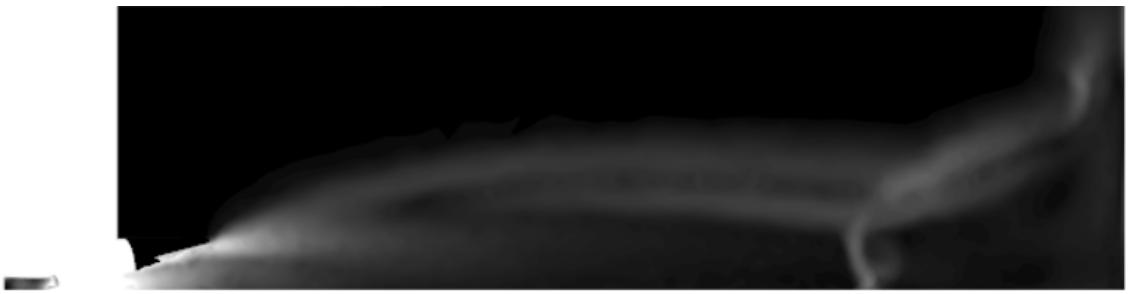


Figure 39: Supersonic Fluidic Injector Simulation: Algorithmically-Obtained Pseudo-Schlieren Image of Density Change, No Injector Flow

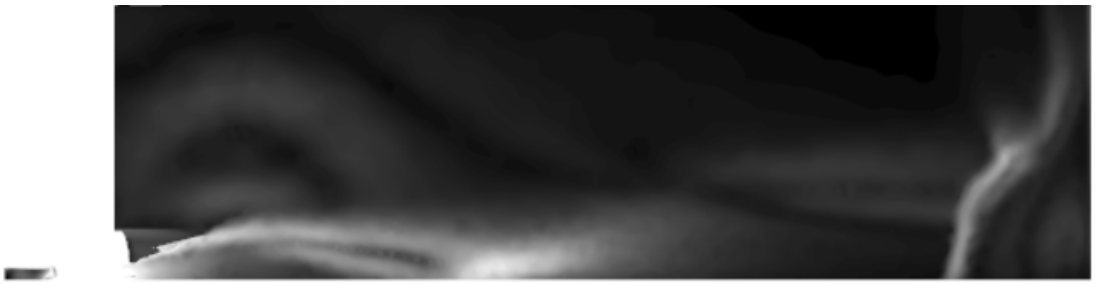


Figure 40: Supersonic Fluidic Injector Simulation: Algorithmically-Obtained Pseudo-Schlieren Image of Density Change, Canted Injector Flow